

Philosophical Problems

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Introduction

Philosophical ideas can, as the history of philosophy shows, have a twofold significance. They can be attempts at the formulation, the treatment, or the solution of certain problems, and they can stand as symptoms of certain tendencies in the intellectual life of man. There is a constant interaction between these two sides of philosophy, in that it may stand as a symptom of a distinctive intellectual movement that a problem is formulated and treated in a particular way, while on the other hand the sting of the problems can release intellectual movements that would otherwise never come to be. There is expressed in this interaction an inner connection between personal life and scientific inquiry, one which asserts itself more or less in every scientific field, but which in the nature of the case will come forward quite particularly as we approach the border regions of the sciences. It comes forward more in the human sciences than in natural science, and it comes forward most of all in philosophy, whose problems are essentially border problems.

It is admittedly no uncommon assumption that there is a conflict between personality and scientific inquiry — that one had best divest oneself of one's personality when one wishes to think scientifically, and that on the other hand scientific methods and results signify nothing for personal life in its free development. I myself in my youth paid homage to such an assumption, under the overwhelming influence of S. Kierkegaard's ideas. That I have come more and more away from it is, I venture to say, because I have learnt to know both science and personal life better.

The need for scientific inquiry is a special form of the need for accord with oneself under all the manifold and shifting experiences. It expresses itself both in the formal sciences and in the real — both as a need to form series of thoughts in which the one member develops with inner necessity out of the other, and as a need to connect the experiences before us in such a way that as comprehensive and as firm a continuity as possible is attained. Personality consists first and foremost in an inner unity and connection among all representations, feelings, and strivings. It is then no sacrifice that personality makes when it stakes its life upon inquiry. It brings forth in science

— as in art — an objective counter-image of itself. Only thus is the inward character that the joy of knowledge can take on to be understood, the *amor intellectualis* that the inquirer feels in working his way through to a standpoint from which the particular can be seen as a member of a great connection. The sacred fire which ever anew sets thought in motion, in spite of all that can damp and hinder it, is understood only when one has this in view. And personality on its side needs to be purified by bowing before the objective connection of thought, the fixed order of things, within which every single being has its determinate place. Freedom is won through truth. It belongs to the nobility of personality that it can by its own power find the great laws which condition its subsistence, determine the wishes that arise in it, and determine the conditions under which those wishes can be realized. There is a science of personal life itself, as there is of everything else; this science is perhaps more imperfect than other science, but it is presupposed in every serious discussion of the forms and demands of personal life. Even Charles Renouvier, the philosopher now living who lays the greatest weight on the opposition between necessity of thought and personality, nevertheless requires “a rational concept of personality”¹ and thus cannot reject every analogy between them.

Such an analogy will come clearly forward when one attempts to formulate the principal problems with which philosophical inquiry occupies itself. These problems are posed at one and the same time from the side of personality and from the side of science.

By a purely historical route I have sought, in my work *Den nyere Filosofis Historie*, to show that there are four such principal problems, namely I) the problem of the nature of conscious life (the psychological problem), II) the problem of the validity of knowledge (the logical problem), III) the problem of the nature of existence (the cosmological problem), and IV) the problem of valuation (the ethical-religious problem).² It will now be my task in this treatise to show the inner connection among these problems. It is one and the same fundamental problem that appears in them in different forms and applications.

It is very different motives that can lead to philosophical inquiry. — Very often it is an ethical-religious, and so a practically personal, interest that leads to philosophy. One then begins with the fourth problem. But it soon shows itself that the treatment of that problem requires insight into the nature of conscious life, into the conditions of knowledge, and into the constitution of the existence in which the valuing personality

¹Ch. Renouvier: *Les dilemmes de la métaphysique pure*. Paris 1901, p. 202.

²These four problems contain the ancient tripartition into Logic, Physics [i.e. Cosmology] and Ethics, deriving from Plato's school (according to *Sextus Empiricus: Adv. mathematicos* VII, 16, from Xenocrates), with the addition of the psychological problem, which in more recent times has pushed itself forward as an independent point of departure.

exists. — If it is an interest in observation that lies at the ground, one will begin with the psychological problem, as has happened in empirical philosophy in its various forms. — If on the other hand it is a need to distinguish between what can be known and what cannot be known, one will begin with the problem of knowledge; this is the road critical philosophy takes. — If one has full confidence in the power of thought and seeks a completed world-view, one will — like the dogmatic and speculative tendencies — begin with the problem of existence. — It will of course not be a matter of indifference, for the kind and manner of the treatment of the problems, which problem one begins with; the one problem will easily be put in the shade by the other. For a comparative doctrine of problems, such as I here seek to contribute to, what matters is to let each single problem come into its full right. I proceed, then, in this way: in keeping with the analogy indicated above between personality and science, I first bring forward the problem of the nature of personal life, that is, the psychological problem, and then pass over to investigating the nature of scientific knowledge, that is, the problem of knowledge. When the problem of existence is placed third in the series, the transition to it is natural both from personality, which stands as a single part of the whole of existence, and from science, whose task it is to lead to a world-view. The three problems named so far would be posed even if man were a purely intellectual, merely knowing being. The fourth problem is posed on account of the relation in which man, as a feeling and willing being, stands to existence. Thus the principal problems of thought hang together with man's theoretical and practical interests.

But more important than the question of the order in which the problems are presented is the question whether they can be led back to one pervading fundamental problem. The possibility of this I find in the significance which the relation between continuity and discontinuity has within each of the several problems. This relation expresses the deepest interest both of personality and of science. In both domains there is a striving, as already remarked, after unity and connection, and the discontinuous stands so far as an obstacle to be overcome. But on the other hand it is discontinuity (difference in time, in degree and in place, difference of quality, difference of individuality) that everywhere, in the domain of science as in that of life, brings new content, releases the bound forces, and sets the great tasks. Neither of the two elements stands, then, as from the outset the only one justified. It will have its unquestionable interest to follow the relation between them under the four different points of view that our four principal problems indicate.

In the philosophy of the nineteenth century the significance of the problem of

continuity comes forward in a characteristic way, in that the different tendencies plead the cause of continuity by turns. In the first half of the century philosophical idealism asserted in its fashion the continuity in existence and looked down on the science of experience on account of the latter's fragmentary character, while positivism (as Comte and Stuart Mill maintained it) impressed the discontinuity between the different groups of phenomena. At the century's close it is realism that, with the help of the hypothesis of evolution, would assert continuity, while the idealist tendency is inclined to lay weight on the unavoidable discontinuity in our knowledge.³ Thus the tendencies change places in the great single combat in which the fight for truth is to be waged. When a significant thought can no longer be carried through from one point of view, inquiry involuntarily seeks a new point of view, and thought gains in clarity through these shifting points of view. The different tendencies relieve one another, like the runners at the ancient torch festival, but the torch is the same. And if none of the tendencies that have so far come forward should have done full justice to the fundamental thought in all the philosophical problems, there is so much the more reason to work at forming it with greater clarity.

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³Compare, besides the work of Renouvier's named above (Note 1), the interesting writings of *Emile Boutroux: De la contingence des lois de la nature*. 2. éd. Paris 1895, and *De l'idée de loi naturelle dans la science et la philosophie contemporaines*. Paris 1895. — Related is the standpoint of *Rudolph Eucken (Der Kampf um einen geistigen Lebensinhalt*. Leipzig 1896).

I. The Problem of Consciousness

1. When we begin the discussion of the philosophical problems by investigating the concept of personality itself, we begin with psychology, and it is then first of all a matter of determining psychology's relation to philosophy. We can here no doubt leave aside the view, which HERBART and LOTZE still maintained, that psychology should be dependent on the view one arrives at concerning the problem of existence, and so dependent on "metaphysics" or cosmology. On all sides an effort is at work to assert the independence of psychology. But even where this effort prevails, there have still been extremely different views about psychology's position.

There are those who hold that psychology is one with philosophy as a whole, and that epistemology, metaphysics, aesthetics and ethics are its different parts. Such a standpoint was taken in earlier times by FRIES and BENEKE, and is taken in our own day by TH. LIPPS.⁴ This standpoint is untenable, partly because personal life is only one of the subjects that can be an object of philosophizing, partly because psychology itself, like all other science, presupposes the general forms and principles of knowledge. The problem of consciousness thus necessarily points beyond itself to the problem of existence and the problem of knowledge, and for all its independence psychology is only one discipline among several. Psychology can then only be a part of philosophy, not constitute the whole of philosophy. It is legitimate to begin with psychology, since psychology describes the place, and the presuppositions, from which we seek to orient ourselves in existence.

But does psychology belong to philosophy at all? Is it not a special science which in and for itself has no more to do with philosophy than natural science or history has? The special roads along which psychological knowledge is sought in the most recent period — the experimental, the physiological and the historical method — do after all seem to testify that psychology is now in the course of becoming a special science and must be separated off from philosophy! — To this the answer is, so far as I can see, that psychology,

⁴TH. LIPPS: *Psychologie, Wissenschaft und Leben*. München 1901. On Fries and Beneke see my work *Den nyere Filosofis Historie* II, p. 221 f.; 236; 240.

for all its special methods, always presupposes the capacity for self-observation. It is subjective, descriptive and analysing psychology that sets the tasks which can then be treated in detail with the help of the special methods. There is here a foundation common to psychology as a special science and to philosophy. Philosophy must build on a general concept of the nature of conscious life if it is to be able to treat the problems of knowledge, of existence and of valuation. It is personality's position as knowing and valuing, and as a part of the whole of existence, that these problems concern, and a concept of personality is presupposed which none of those special methods is able to furnish. They investigate single expressions of conscious life, but their investigations must be worked together if a concept of personality is to be formed.

But here precisely the fundamental problem arises in the psychological domain: is it possible by the road of experience to arrive at a concept of personality? Does our conscious life form a whole, a continuum, a little world of its own, or should it be merely an aggregate, a sum of elements and fragments? This is the question that philosophical psychology (or, if one prefers, the philosophy of psychology) raises. And in its treatment of it it relies partly on self-observation and descriptive psychology, partly on the results of the experimental, the physiological and the historical method.

Experience shows us greater and lesser interruptions of our conscious life; there are unconscious intervals between the conscious states. And when we investigate the single state of consciousness more closely, we can distinguish within it different elements which can be found again in other states, so that the single state, no less than consciousness as a whole, might seem to be a resultant of a manifold of elements or fragments which stand as what is properly real. Consciousness would then be an aggregate, or a sum of single flashes. A psychical continuum, such as is presupposed in the concept of personality, there would not be.

To this it must first be remarked that it will never be possible to decide with full certainty whether analysis has reached down to the last elements. The possibility always stands open that the elements at which we stop, and out of which we are inclined to think consciousness composed, are themselves composite, and that if we could reach down to still simpler elements, elements of the second or third order, a continuity would appear which could not be exhibited so long as we stopped at the elements of the first order. Our sensations, which one has been inclined to regard as a kind of psychical atoms, nevertheless bear — as recent investigations along various roads have shown — traces of processes of composition of a still more elementary kind than those which

observation shows us more directly.⁵

This, however, is only a preliminary consideration, and leads to no settled result, since the defender of discontinuity may very well use it to show that all the continuity that can be exhibited is only preliminary. What is on the other hand of decisive importance is that the so-called psychical elements are always determined by the connection in which they occur, and that it is only by abstraction that one ascribes to them, outside this connection, properties which they have only within the connection itself. It is on this fundamental thought that my presentation of psychology is built. All mental life works in the manner of genius: it brings forth immediately and involuntarily connected phenomena and processes which analysis afterwards seeks, in a more or less artificial way, to dissolve into “elements”. As regards sensation we can here appeal to Fechner’s law and to the law of contrast, according to which the degree of intensity and the quality of every single sensation are determined by the whole connection in which it comes to be. The connection cannot, then, be conceived as a product of the psychical elements, since these come to be as *these* determinate sensations only in virtue of the connection. Did they occur in another connection, they would not be the same sensations. The relation is analogous as regards representations. The association of representations finds its ultimate explanation in the unnaturalness of isolating the single representations; there is always at work — and to a higher degree, the more vigorous consciousness is — a tendency to supplementation or extension, whereby the single representation enters into combination with other representations according to determinate laws. The so-called “Association-Psychology” (among whose adherents I have myself sometimes wrongly been reckoned) conceived the single representations as independent atoms which were brought into connection with one another in a purely external, mechanical way. But it is precisely the reverse: it is the connection of the whole that in the process of association exercises its power over the single representations, which moreover never occur in a wholly isolated condition, such that their combination could stand as a mechanical product. Analysis here too always presupposes synthesis. In thinking proper this relation shows itself plainly when we compare the formation of judgments and the formation of concepts. These mutually presuppose one another, since our judgments, which are after all combinations of concepts, can only be complete when the concepts combined in them are complete, while conversely the formation of concepts presupposes a series of judgments by which different elements are determined in their mutual relation. It shows itself here again that the whole and the parts mutually determine one another. There are

⁵See my *Psychologi* V A, 2.

no wholly isolated concepts which could only afterwards be combined into judgments.

There expresses itself here an antinomy which hangs closely together with the essence of consciousness and is peculiar to the concept of personality. Consciousness and personality cannot be explained as products of antecedently given elements, any more than organic life can be explained as a product of inorganic elements. But on the other hand the essence of consciousness and of personality, like the essence of organic life, displays itself in a constant gathering-together of *given* elements, elements which they do not themselves originally bring forth. It is this antinomy that makes the coming-into-being of life and of personality such great riddles.⁶

In the foregoing, account has been taken only of the more formal connection within conscious life. But in every consciousness there is a goal towards which there is striving, a guiding interest which takes, or strives to take, everything into its service. This guiding interest – the fundamental value, one might call it – can change in the different ages of life; but there will always anew be a tendency to the development of such a one, and in higher or lower degree it sets its stamp on all the elements of consciousness, gives them their direction. It constitutes the soul proper (when by “spirit” we think chiefly of the formal connection).

It is the relations here brought forward that make the concept of personality have always to stand as the central thought in psychology. When analysis and the special methods make their sections and seek to isolate single elements or moments, that is legitimate, just as it is legitimate to attempt to determine an irrational number by adding decimal to decimal, as far as one can. But the irrational relation between the whole and the elements continues to subsist.

From the idealist side one has often been inclined to take the concept of personality for something finished and to operate with it in cosmological speculations. What is thereby overlooked – and it is what the positivist tendency particularly emphasizes – is that we are constantly at work forming the concept of personality, spelling our way forward to a psychological conception of the whole, just as biology is still constantly spelling out a definition of life. And just as biology, for all its recognition of the peculiarity of the living organism, knows no other method than that which through observation, experiment and analysis seeks to understand the composite processes by the help of the simpler, so psychology, however firmly it holds to the synthetic character of conscious life, can apply no other methods than those which all science applies: observation, experiment and analysis. The concept of personality stands as the ideal towards which the course is

⁶Compare my treatise *Det psykologiske Grundlag for logiske Domme* (Vid. Selsk. Skr. 6. Række), p. 16 f.; 59, and my treatise *Om Vitalisme* (reprinted in “Mindre Arbejder”).

set, the standing problem which all the special methods serve to illuminate.⁷

The irrational is here, as everywhere, not merely a barrier but also a task, a task that is set ever anew. Descriptive psychology, which leads one particularly to lay weight on the connection of the whole in and with which mental phenomena appear, will here — especially as regards the higher or more developed phenomena of consciousness — always preserve its independence and its importance in relation to experimental psychology; the latter in the end always receives its tasks from the former. And descriptive psychology serves at the same time as a corrective to experimental psychology, which by its method easily comes to isolate single elements too much, to overlook what is purely involuntary in conscious life, and to lay too great weight on the outward symptoms of states.⁸ On the other hand descriptive psychology will be able to draw advantage from experimental psychology, in that what the latter is able to exhibit concerning the more elementary psychological processes can by the road of analogy throw light on the nature of the higher processes and give description greater fullness and accuracy.

Descriptive psychology approaches art and becomes in the end an art; experimental psychology approaches natural science. But there is nevertheless no difference of principle between them. There is not, as MÜNSTERBERG in the most recent period especially has wished to maintain,⁹ a gulf between them. The former is to be a “science of value”, the latter a natural science; the former is to operate with the “concept of freedom”, the latter with the concept of cause. By freedom Münsterberg understands the capacity to act according to purposes. But should then the psychical process by which a human being comes to set himself a purpose and to work at attaining it not be an object for scientific psychology? And can it be understood otherwise than by being investigated in connection with the simpler processes that take place when pleasure and pain, joy and sorrow arise, or when representations are combined and separated? Here there is a mass of experience to be gathered and collected; what matters is to follow a development step by step, to establish the different degrees of consciousness and the differing impulses and motives that co-operate. There will further be a distinction to be drawn between the different individual types that can appear with respect to the kind and direction of the life of will, and a comparative individual psychology will be able to serve as a

⁷Of interest for the standpoint of modern positivism on this problem, in contrast to the standpoint Comte and Stuart Mill took, is ROBERTO ARDIGÒ's work *L'unità della coscienza*. Padova 1898.

⁸Compare my *Psychologi* I, 8 d.

⁹MÜNSTERBERG: *Psychology and Life*. Westminster 1899. — Münsterberg allies himself with the view that Windelband and Rickert have urged, according to which there is a gulf between natural science (to which psychology too is reckoned) and the science of culture. On this view see the apt remarks in WUNDT: *Einleitung in die Philosophie*. Leipzig 1901, p. 65–74, and in GUIDO VILLA: *Psychology and History*. (The Monist, January 1902.)

necessary aid to understanding the development within the consciousness of the single person. It is then a wholly unnatural boundary that is drawn when one lays down a difference of principle between “personal” and “psychophysical” categories — and especially when this opposition is made the ground of an opposition between “the truth of life” and “the truth of science”.¹⁰ It is after all the task of science to understand life, even if complete understanding is always an ideal, and it is life that yields science the material of experience on which it builds. Here there is a constant relation of interaction, but no absolute opposition. And there are many intermediate links and intermediate forms between science’s separating out of the single elements of existence and life’s great interplay of all elements, which in the end only art can present in its whole fullness.

2. Nevertheless great difficulties always remain for the scientific treatment of the phenomena of consciousness. To a real understanding it belongs that a continuity be exhibited in the processes one investigates. It would be psychology’s ideal not only that we could win so complete a description of the states of consciousness that each single one of them came to stand as a distinctive member of the whole psychical process, but also that we could reduce the differences between the changing states to forms so simple that the following appeared as continuations, or as transformations, of the preceding. But the prospect of reaching this ideal is barred by the discontinuity which experience seems to establish. There are, after all, unconscious intervals between our conscious states — in fainting, in dreamless sleep (so far as there is such a thing); and there are qualitative differences between the different states and elements of consciousness, so that every single state and every single element seems to come to be out of nothing, if we keep to the strictly psychological consideration. Between the different individual consciousnesses there is in any case a great and conspicuous discontinuity. The one consciousness can be derived from the other still less than the one state of consciousness can be derived from the other.

These psychical relations of discontinuity come forward especially sharply when they are set over against the continuity, and the relations of equivalence, which the material phenomena present and of which men became aware very early. Among the Greek natural philosophers the persistence of matter through all changes was a common assumption, and this assumption has received its confirmation in the chemistry of more recent times. In sharp contrast to it stood the distance between souls among themselves.

¹⁰MÜNSTERBERG, *op. cit.* p. 282: “If we force the system of science upon the real life, claiming that our life is really a psychophysical phenomenon, we are under the illusion of psychologism. If, on the other hand, we force the views of the real life, the personal categories, upon the scientific psychophysical phenomena, we are under the illusion of mysticism. The result on both are the same. We lose the truth of life and the truth of science.”

When DESCARTES applied the concept of substance both in the psychical and in the physical domain, it was with the great difference that he assumed many soul-substances but only one single material substance.¹¹ Thereby the psychical discontinuity was pronounced sharply and definitely, in contrast to the physical continuity. It is only in a half or wholly mystical fashion that we find, in the history of philosophy, the continuity of mental life asserted in such a way that the single consciousnesses are thought to stand in the same relation to a universal soul-substance in which the single bodies stand to the universal material substance. So with AVERROES, SPINOZA and HEGEL.¹² In the most recent period it is now not so much the relation of discontinuity between individual consciousnesses as the relation of discontinuity within the single consciousness that draws attention to itself. And there are in particular two considerations which have here led to the problem's being posed more sharply than before.

Physiology has come to full consciousness of its method and of its independence as a science, especially after the reform it underwent about the middle of the nineteenth century. It now consistently requires that every state of the brain be explained from the preceding states of the brain and of the organism — that physiological phenomena in general be explained by physiological causes, in so far as they are not effects from the physical world outside, continued into the organism. On strict natural-scientific method every material state within and outside the organism must find its explanation by being seen as having arisen through the conversion, into a new form, of the energy which expressed itself in the preceding states. When two brain-states succeed one another, the explanation will accordingly first have been reached when it is shown that all the energy which expressed itself in the first state has been converted without remainder into nutritive activity, heat, electricity, motor impulses and so on. It is indeed difficult to

¹¹DESCARTES (*Synopsis meditationum*) teaches that all substances are imperishable, but that it is only *corpus in genere sumptum* that is substance, not for example *corpus humanum*; every soul, by contrast, is *pura substantia*, which does not change, however much its single *accidentia* (thoughts, feelings and expressions of will) may vary. — In his work *Cogitata metaphysica* (II c. 11) SPINOZA says (from the Cartesian standpoint which in that work he essentially adopts): *Si ad totam Naturam materiæ attendamus, illi nihil novi accedit; at respectu rerum particularium aliquo modo potest dici, illi aliquid novi accedere. Qvod an etiam locum habet in rebus spiritualibus, non videtur: nam illa [i.e. spiritualia] ab invicem ita dependere non apparet* [souls do not depend upon one another as bodies depend upon one another]. Later Spinoza came to another view.

¹²Relying on his interpretation of Aristotle's "active reason", AVERROES (on him see RENAN: *Averroës et l'Averroïsme*. Paris 1852) teaches that while the single souls arise and perish, there subsists *intellectus universalis*, the world-thought that is at work in the thinking of the single souls. In the Middle Ages this doctrine was renewed, among others, by SIGER OF BRABANT in his *Qvæstiones de anima intellectiva*, recently edited by Mandonnet. (Mandonnet: *Siger de Brabant et l'Averroïsme latin au 13^e siècle*. Fribourg 1899.) The whole doctrine of the actual subsistence of spirit goes back to Neoplatonism. (See PLOTINOS: *Ennead.* V, 9, 6.) — On SPINOZA's doctrine of *intellectus infinitus* and *idea dei* see *Den nyere Filosofis Historie* I, p. 290; on HEGEL ib. II, p. 165.

imagine such an experiment carried out; but it would mark the carrying through of the physiological method. Neo-vitalism itself knows no other method for the investigation of physiological states. Thus is confirmed SPINOZA's saying, that when one appeals to the intervention of the soul in explanation of a bodily state, it is the same as saying that one does not know what the cause is. Even if an experiment should show that at the arising of an organic state energy arose or disappeared without physical equivalent, one would surely sooner believe that there was a fault in the experiment than acquiesce in the result. "The principle of the conservation of energy has", says MAXWELL,¹³ "acquired so great a scientific weight that no physiologist would have any confidence in an experiment which showed a considerable difference between the work performed by a living being and the reckoning of energy received and expended." — By carrying through the physiological method, understood in this spirit, an ever greater continuity will be exhibited in the organic processes to which the phenomena of consciousness in fact show themselves to be attached. Physiology is therefore far more happily placed with respect to the principle of continuity than psychology will ever be able to be, and it might therefore seem a well-founded demand that CARL LANGE put forward when he declared that all psychological definitions ought to be replaced by physiological ones.¹⁴ The psychical phenomena then come to stand as mere provisional symptoms.

The second consideration attaches to the one just adduced, but is of a general epistemological kind. It rests on the proposition that a complete understanding is won only where the relation between cause and effect can be made a relation of identity or of continuity, such that a quantitative equation becomes possible between the phenomena that are connected in the causal relation. It is the ideal concept of cause that is here maintained, in contrast to the empirical concept of cause, by which phenomena that are qualitatively different stand as connected in unavoidable succession. The closer investigation of these two concepts of cause I postpone to the problem of knowledge. Here it need only be remarked that there is no doubt that where the ideal concept of cause can be carried through we obtain a more complete understanding than where we must stop at the empirical concept of cause — and have, perhaps, difficulty enough even

¹³MAXWELL: *Scientific papers*. Vol. II. Cambridge 1890, p. 759. Shortly before the passage quoted, Maxwell remarks that whereas the thought that the soul might be found somewhere in the brain by anatomical means has nowadays as a rule been given up, the thought has held on longer that by pursuing the material processes far enough back one will come to a material process brought about by the soul. It is against this possibility that the quoted passage is directed. Compare SPINOZA: *Ethica* III, 2, Schol. — His conception of the principle of inertia in his work *Matter and Motion* § 41 leads to a result similar to Maxwell's conception of the principle of energy. — Compare my *Psykologi*⁴ p. 30 and my *Psykologiske Undersøgelser*, p. 82–86.

¹⁴CARL LANGE: *Nydelsernes Fysiologi*. København 1899, p. 45.

in applying that. Now both the material and the psychical phenomena present qualitative differences; but in the material domain it is possible to conceive these qualitative differences as differences of quantity, namely by virtue of the law of the conservation of energy and of matter, while nothing corresponding is possible in the psychical domain. The right psychological method is therefore taken to consist in substituting the physiological phenomena for the corresponding psychical ones; that is the only way in which we can put quantitative determinations in the place of qualitative ones and thereby carry through a stringent causal connection. This conception of the psychological problem has been carried through most strictly by RICHARD AVENARIUS in his acute work *Kritik der reinen Erfahrung*. He requires there, as the condition of full scientific understanding, a reduction of “the dependent vital series”, by which are meant the psychical states, to “the independent vital series”, the corresponding physiological states, whereby alone it becomes possible to reduce the empirical differences to the least possible, or, as Avenarius expresses it, to reach “a heterotic minimum”. In a similar direction goes MÜNSTERBERG’S view, as he has expressed it in his work *Psychology and Life*. His principal contention is that since psychical phenomena are not quantitative, they cannot be members of a causal series.¹⁵

As against the views which thus require a reduction of psychology to physiology in order that a scientific psychology should become possible — which really, then, would abolish psychology in order to make it a science — it must be acknowledged that they can appeal to the factual discontinuity and the qualitative difference between psychical phenomena, which will always more or less set the limit to a strict carrying through of psychology as a science. Whether the road they point out leads to the goal, and whether the method they commend does not contain a palpable self-contradiction, is another matter.

When it is required that the psychological definitions be *replaced* by physiological ones, it is clear that the presupposition of this is that the psychological definitions must first be finished. But to procure those definitions must be psychology’s business, and if it cannot procure definite and secure definitions, physiology cannot know what it is that it is to seek an explanation of in the brain: if what is to be replaced is vague and wavering, the replacement will be so accordingly. There will be no security that one has really found the brain-states which correspond to the psychical phenomena one has in view. Psychology’s independence must therefore in any case be acknowledged, since it is psychology that here — as a kind of symptomatology — is to set physiology

¹⁵MÜNSTERBERG: *Psychology and Life*. p. 127: “Mental facts, as they are not quantitative, cannot enter into any causal equation.”

its tasks. Now for every science of experience it is a long and difficult labour to find conclusive definitions; such are possible only when the science of experience is in essentials completed; they come at the end, not at the beginning, of the investigation. Far too often one operates with unfinished psychological definitions which are regarded as trustworthy points of departure for the physiology of the brain. Thus DESCARTES, and in more recent times LOTZE, regarded the very concept “soul” as so distinct and clear that one could confidently invite physiologists and anatomists to search for “the seat of the soul”. In the most recent period FLECHSIG¹⁶ has regarded the concept “association” as so simple and clear, and withal as so independent, that he has been able to think he could find a peculiar place in the brain for the functions that concept designates. That it is nevertheless a highly imperfect concept of association that Flechsig here takes as his ground is seen from the fact that psychological experience never shows us absolutely single psychical elements which are only afterwards to be brought into connection with one another by a peculiar process. One cannot set the process of association in definite contrast to the processes by which the single psychical elements (sensations and representations) come to be. It is a psychological abstraction, not a real psychological definition, that Flechsig operates with. Anatomical objections have been raised from several sides against Flechsig’s doctrine of peculiar association-centres, which I cannot judge; but the psychologically unfinished character of his concept of association shows sufficiently how difficult a task it is to “replace” psychological definitions by physiological ones.¹⁷ A principal point for the criticism of Flechsig’s theory seems to me to be the following circumstance. When there are psychical elements which do not enter into such an association as might under usual conditions be expected, then on closer investigation the cause will show itself to be that the elements in question are each of them members of other, fixed associative relations from which they cannot be released. It will in any case always, psychologically regarded, stand as a task to find the preceding associations which hinder later associations that are in and for themselves “natural”. By isolating the two concepts element and association from one another one will thus get both a false psychology and a false physiology. For an exhaustive definition of a concept like association long investigations will still be needed, and physiology will have to wait a long time if it means to “replace” a psychological reality and not a mere abstraction.

But let us suppose that the psychological definitions are finished and complete —

¹⁶FLECHSIG: *Gehirn und Seele* 2. Leipzig 1896, p. 24 f.

¹⁷Compare the criticism of Flechsig’s theory in O. VOGT (*Flechsig’s Associationscentrenlehre, ihre Anhänger und Gegner. Zeitschrift für Hypnotismus*. V, 6) and ALB. ADAMKIEWICZ (*Die Grosshirnrinde als Organ der Seele*. Wiesbaden 1902, p. 75 f.).

will not the greatest problems still remain unsolved? How can physiological states have psychical symptoms? How can qualitative differences correspond to quantitative ones, and how can discontinuous phenomena be attached to continuous processes? Far from such questions falling away with the reduction of psychology to physiology, they are precisely sharpened by it and appear in a more glaring form. And in any case it stands as an unsolved riddle how the qualitative differences and the discontinuity arise. Even if one succeeds in showing that two concepts A and B , which one has hitherto regarded as different, are identical, so that one can set $A = B$, one has not thereby explained how A and B could stand as different in the first place. To call this difference merely “subjective” does not help; it is not thereby got out of existence — it is precisely the fact that existence has a subjective side as well that makes psychology possible and necessary. And if one declares (with MÜNSTERBERG) that these subjective phenomena can neither be described nor explained,¹⁸ it is not easy to understand how one is to get anything to put into a “causal equation”. One then behaves like the man who saws through the branch he is sitting on.

It is inconsistent to deny a causal relation between psychical phenomena among themselves while assuming a causal relation between the corresponding physiological states. If one acknowledges a physiological lawfulness, one must acknowledge a psychological lawfulness also. If the psychical phenomenon α corresponds to the brain-state A , the psychical phenomenon β to the brain-state B , and if there is a causal relation between A and B , then a causal relation — in the sense of unavoidable succession — must appear in psychological experience between α and β as well. And — as has been shown above — it is in reality this causal relation between α and β that leads one to look for a causal relation between A and B . In AVENARIUS’s strict carrying through of the reduction of psychology (the doctrine of “the dependent vital series”) to physiology (the doctrine of “the independent vital series”) it is likewise plain at every point that it is from the “dependent vital series” that he infers the constitution of the “independent” one. For the most part what is constructed is only a physiological schematism, offered as an “intuitive” presentation of what goes on in the psychical processes. Neither physiology nor psychology is yet so complete that one can get beyond such a schematism.

The psychical phenomena do not, however, present so great a discontinuity and such purely qualitative differences as is often supposed. Exact observation often leads to the discovery that where there seemed to be a psychically empty space there was in reality a psychical content given, although it was not grasped by attention or by recognition,

¹⁸MÜNSTERBERG: *Psychology and Life*. p. 172: “The reality of the will and feeling and judgment do not belong to the describable world.” — p. 208: “The subjective attitude is never object; it is never perceived.”

or although it was forgotten immediately after being experienced. And as regards the qualitative differences, here too more exact observation will find more and finer nuances than those at which one first stopped and which one perhaps confidently declared to be wholly disparate. The continuity in conscious life can therefore be carried further than is often assumed, and there can be a scientific danger in proclaiming too strongly that difference of quality and discontinuity are the distinguishing mark of conscious life. The inner connection in the content of consciousness is a fact as well, and functions such as memory and comparison are quite as significant for the characterization of conscious life as are the phenomena that bear the stamp of discontinuity. Psychology's task therefore naturally becomes that of exhibiting, so far as possible, the connection and combination of the single elements, so that the whole is understood through the parts and the parts through their relation to the whole. That this leads to an antinomy which in the end makes the psychological problem insoluble for us, we have already seen. But there is work enough to do before we come to that limit (which moreover cannot be proved to be absolute). LEIBNIZ has the merit of having energetically maintained the principle of continuity in the psychical as well as in the physical domain, especially by pointing to those psychical elements which — in relation to the clearly conscious states — are vanishingly small, and which only by being summed and combined bring forth a result accessible to unpractised self-observation. A comparison between modern and ancient or medieval poetry alone shows how many more mental nuances self-observation has now caught sight of; it is hardly only because differentiation in the psychical domain is now greater than in earlier times that this greater richness is found. Knowledge and understanding of what is immediate and involuntary in mental life has increased above all. Childlike and primitive psychology has a tendency to conceive everything that goes on in the soul as clearly conscious and as resting on deliberation and intention. In modern psychology the burden of proof — in practice as in theory — has been shifted: it now falls on the one who asserts that an action has been performed with deliberation and express decision. And the transition between the unconscious and the conscious, and within the conscious between the involuntary and the voluntary, itself takes place continuously; the more practised self-observation and criticism are, the more difficult it becomes to lay a finger on one definite point at which the boundary is supposed to lie.

Leibniz was inclined to reduce all psychical differences to differences of degree in respect of clearness and obscurity.¹⁹ This attempt was characteristic of the intellectualism of the age of Enlightenment. But there are other ways than such a reduction in which

¹⁹*Den nyere Filosofis Historie* I, p. 334–336.

empirical psychology can conceive a continuous connection despite the qualitative differences which the psychical processes present at their different stages. Fusions and compositions take place at the different levels of conscious life, from the simplest sensations to the highest states of feeling. And displacements of motive take place, by which what at first has only mediate value later acquires immediate value, or conversely. Finally, conscious life in wholly integrated characters — psychology's highest and noblest objects — shows such a dependence of everything that stirs in the soul upon one purpose, one guiding thought, that we here find a causal connection quite as firm and as intimate as any in the physical domain. Every single thought, every single mood and every single impulse is in such a character plainly determined by the whole to which it belongs, and which it itself helps to form.

Cases enough remain in which psychical continuity cannot be exhibited. The question is whether we are therefore entitled to deny it. Apart from what continued observation may establish, the great question arises here whether psychical phenomena can come to be from material causes, when the concept of the material is conceived in the way natural science has hitherto conceived it. If the natural-scientific concept of the material includes no possibility of an arising of psychical phenomena, then it will be fully legitimate to set up the concept of potential psychical energy, in order to mark that we do not mean to let go of the principle of continuity even where we cannot apply it in detail in experience. It is indeed for a similar reason that the concept of potential energy has been introduced in physics, despite the obscurity that attaches to it. Everywhere it contains an acknowledgment of a limit at which one nevertheless does not mean to give up the connection one has hitherto found. The mere fact that psychical elements can be reproduced after an interval in which they have not been in consciousness already necessitates the use of concepts like "disposition", "trace", "possibility" and the like, which in reality express the same thing as is meant by "potential energy". That in applying this concept in the physical domain we are more happily placed than in the psychical²⁰ does not exclude the legitimacy and the necessity of setting it up here. One can of course, in a purely descriptive presentation, content oneself with stating the different psychical phenomena that emerge at certain moments, often without observation finding any psychical connection between them. The psychical phenomena then stand as scattered flashes, in strong contrast to the continuous connection which the corresponding physiological phenomena present. And since what appears with the highest degree of continuity easily comes to stand for us as more real than what is

²⁰Compare my *Psychologi* V B, 6.

discontinuous and momentary, the physiological phenomena will easily come to stand as what is properly real, the phenomena proper, and the psychical phenomena will come to stand as something superfluous, an accidental plus, or, as it has been expressed, as epiphenomena. Such an “epiphenomenalism” has surely never been set up as a theory proper; it is merely an empirical registering of what is enigmatic in the appearance of the psychical phenomena as compared with the more closed continuity in the series of material phenomena. Nor indeed does it furnish any explanation; on the contrary it abolishes both every psychological and every physiological explanation, if it means to be more than a mere description. And even as description it easily becomes precipitate, since it easily comes to stop too early at what is discontinuous, without investigating whether there are not more transitions and nuances to be found than have hitherto shown themselves. The concept of potential psychical energy will here — precisely by being really only the expression of an unsolved task — contain a standing summons to go further in observation and analysis, in order if possible to discover the reality in which the “possibility” in the end always consists.

3. When weight is laid, from the psychological and the physiological side alike, on preserving continuity in as high a degree as possible, the identity hypothesis becomes the real working hypothesis for the problem of the psychical and the physiological. Like every significant hypothesis it is really only the expression of a method. We have in the psychical and the physiological phenomena two series of states, or forms of state, which empirically vary in a certain relation to one another, without its being possible, however, to *derive the existence* of the one series from the *existence* of the other. Scientifically regarded, the task must then be to conceive the two series each for itself as completely and as continuously as possible, and to show which determinate members of the one series correspond to determinate members of the other. We may rightly regard members of the one series as symptoms of members of the other. Sometimes it is the psychical, sometimes the physiological states that are the more accessible, and from which we therefore take our point of departure. The close belonging-together of the two series of states makes it impossible to refer them to two different “beings” or “things”; it will be natural to conceive them as different forms of expression of one and the same “being”. From the fact that two properties cannot be derived from one another one has no right to infer that they belong to two different things, especially when they vary in a determinate mutual relation. Sulphur’s form of state (solid, liquid and so on) and its colour (yellow, brown and so on) vary in a determinate relation to one another and yet cannot be derived from one another, although one can, when one knows the two series of changes, infer

from the one to the other. In the case of sulphur we do know the common cause of the two series of changes, namely heat; it is heat that expresses itself in these two different ways. In the relation between the psychical and the physiological series the analogous knowledge is denied us. Our experience is here insufficient to solve the problem. This hangs together with the fact that both our knowledge of the psychical and our knowledge of the physical are empirically limited. What matters, then, is to work in both fields of knowledge in such a way that at no point does one arbitrarily cut the thread.

The physiological continuity is both a consequence of the law of the conservation of energy (really already of the principle of inertia, on the strict interpretation of it) and an expression of physiology's independence as a science. It furnishes a fruitful and indispensable principle of research by requiring constant demonstration of physiological causes of, and physiological effects of, physiological states. As we have seen above, the principle we stand at here is so essential that — if an experiment pointed in the opposite direction — one would sooner assume an error in the experiment than a breach of the principle. The principle of continuity — and with it the identity hypothesis — would be refuted if it could be shown that the energy contained in a brain-state did not stand in a relation of equivalence to preceding and succeeding states in the brain, in the organism, and in the physical world outside. It would already be a matter of concern for the identity hypothesis if it could be shown that the psychical phenomena in time go *before* or follow *after* the corresponding physiological states. But in an attempt to establish this it is not enough to show what popular observation already shows, that sensation arises later than the corresponding sense-impression, or earlier than the physiological (motor, and especially vasomotor) expressions which in one way or another hang together with it. For between the psychical states and the peripheral processes in the sense-organs and the organs of movement lie the central physiological processes, which are not so readily accessible to exact and direct experiment. And even if such a direct experiment were possible, it would not be decisive: because the phenomenon *B* follows *A* in time, *A* and *B* may very well be effects of the same cause, which first deposits the effect *A* and then the effect *B*. There might be peculiar causes which made the one vital expression alter more rapidly than the other. — Finally it would be a proof against the identity hypothesis if it could be established that different psychical phenomena could correspond to one and the same physiological state. It has been supposed, for example, that the brain's significance for conscious life might consist in counteracting organic life's tendency to habitual, automatic operation. The brain has a wonderful capacity for initiating new movements which, as soon as they have been practised, become habitual actions that

are left to lower centres to carry out. Only by the brain's help has thought been able to free itself from automatism and to use the organism in its service without sinking under the power of habit. But different thoughts can at different times make use of the same "motor schema", the same suspension of an automatic tendency. This possibility has in the most recent period been advanced by BERGSON as an objection to "strict parallelism".²¹ This objection touches on a question that has come up very little in the frequent discussions of the psychophysical problem. Whether or not one accepts the special theory the French philosopher has set up, it seems clear that the closer one presses the problem to life, the more extraordinarily difficult it will become to find those members of the two series of phenomena that can be said to be "corresponding". Where a qualitative difference is present, one can establish only by roundabout ways which members correspond to one another. When we know how heat acts on a substance's colour and on its form separately, we can determine which changes of colour and of form "correspond to one another". But it is precisely an analogous knowledge that is not at our disposal in the discussion of the psychophysical problem. We are then limited to purely empirical inferences. And such will here be more difficult to draw than Bergson seems to assume. For when sufficient weight is laid on continuity in the psychological and the physiological domain alike, it will prove difficult to resolve the two series into members that appear with such independence and distinctness that a real comparison can take place. But whatever hypothesis we take as our basis, we must be prepared for it that the members of the two series which may be taken to correspond to one another will present differences that could not be inferred from the one member or the other *taken by itself alone*. It might therefore very well be that "different" psychical phenomena could correspond to "the same" physiological state (or conversely), just as one language sometimes has only one word where another has two. It is then the connection that is decisive, and it will surely be decisive to such a degree — in the psychological as in the physiological domain — that it is only with a slight degree of approximation that one will be able to speak of "the same" phenomena or states. "Parallelism" is therefore not touched by that objection, provided we make sufficiently strict demands on the demonstration that the states really are the same.

It is essentially as a working hypothesis, not as a positive solution, that the identity

²¹See the interesting discussion of *Le parallélisme psychophysique et la Métaphysique positive* in Bulletin de la société française de Philosophie. Juin 1901. In the course of that discussion Bergson remarked, among other things (p. 51): *Étant donné un état psychologique, la partie jouable de cet état, celle qui se traduirait par une attitude du corps ou par des actions du corps, est représentée dans le cerveau: le reste en est indépendant et n'a pas d'équivalent cérébral. De sorte qu'à un même état cérébral donné peuvent correspondre bien des états psychologiques différents, mais non pas des états quelconques. Ce sont des états psychologiques qui ont tous en commun le même "schéma moteur"*.

hypothesis (for which “parallelism” and similar expressions are unhappy and misleading designations) has its significance. I for my part have in any case always maintained it as an “empirical formula” which can guide us in our investigations in such a way that neither physiology’s right nor psychology’s is violated by an investigation on the one side or the other being broken off too early. The physiologist may be tempted to stop too early if he can expect at some point or other to come upon “the soul” as the cause of a change in the brain’s state,²² and the psychologist may be tempted to stop too early if he can expect at some point or other to stand before a psychical phenomenon supposed to have its cause in a “nerve process” or in “nerve energy”.

4. It will nevertheless have its interest, from both sides, the psychological and the physiological, that they should approach one another as closely as possible by going back to the fundamental concepts of the two sides under which existence here appears in our experience.

From the psychological side the *concept of will*, taken in the widest sense as the concept of psychical activity, will stand as the fundamental concept. It might seem to tell against this that several psychological writers, precisely in the most recent period, seek to eliminate the concept of will from psychology — not because they deny what popular usage calls will, but because they hold that this concept does not designate so fundamental a point of view as knowledge and feeling do, the so-called phenomena of will being supposedly reducible to elements of knowledge and of feeling in peculiar combinations. In support of this one may appeal to the fact that will as such, our activity as conscious beings, cannot be an object of immediate self-observation in the way that representations and feelings can be.²³ We experience will’s motives and will’s results, but not will itself — just as in the domain of material nature we experience energy’s conditions and expressions, but not energy itself. HUME has already shown this for all causality, and so for psychical and material causality alike. The concept of will is formed, like the concept of energy, by a construction. But it is a construction we are compelled to make. And if one determines the concept of a psychical element, not in such a way that it must necessarily be something that can be an object of immediate self-observation, but in such a way that it designates an essential and irreducible property of conscious life, then will may very well be a psychical element, and the concept of

²²An example of this I have adduced in my *Psykologi*, 4. Udg., p. 67 Note.

²³Compare my *Psykologi* VII B, 4; *Psykologiske Undersøgelser*, p. 67–81; *Søren Kierkegaard som Filosof*, p. 74–81. — EBBINGHAUS: *Grundzüge der Psychologie* I, p. 168 (compare also his article in *Zeitschrift für Psychologie* IX, p. 201), and EHRENFELS: *Werttheorie* I, p. 245–249, infer from the unobservability of will that will is not a special psychical element. Compare my criticism of the last-named work in *Göttinger gel. Anzeigen*. 1900, p. 742 f.

will a fundamental psychological concept. The reason we cannot make will the object of self-observation, as sensations, representations and feelings can be, might lie in its extending through all the changing states and forms of conscious life as their constant presupposition. Consciousness subsists only through an unbroken labour of gathering the single elements into a whole. Such a labour of combination and concentration expresses itself in the simplest sensation as well as in every process of representation, every feeling, every impulse and every decision. At every point there expresses itself an activity which is just as original a side of conscious life as are the elements (sides or properties) that observation and analysis immediately find before them. It is not the case that we first have sensations, representations and feeling, and that then, by their combination, something arises which we can call will. Without an original combination, a primary synthetic process, the very elements which determine will in the narrower sense do not arise at all. In the special expressions of will (reflex action, impulse, wish, intention, decision) the primary capacity of concentration expresses itself in a peculiar way under the influence of certain determinate elements of knowledge and of feeling.²⁴

— There thus takes place a constant interaction between the element of activity, the will-element proper, and the intellectual and emotional elements. We meet here again the antinomy mentioned above (1), which asserts itself in all conscious life, indeed in all life. But the main thing here is that we can form a concept of energy already on purely psychological ground, since psychical work is performed wherever a psychical phenomenon appears, this always — so far as we can trace it — presupposing a synthesis. The psychical work in which the synthesis consists is the greater, the more the single elements are qualitatively different, and the further apart they lie in time.

It might now be tempting to identify this psychical energy straightway with the energy that operates in the nerve tissue. But since not all nerve processes are connected with phenomena of consciousness, a distinction must be drawn between conscious and unconscious nerve energy, and the problem poses itself anew. Besides, we know nothing more precise about this so-called nerve energy, or about its relation to other forms of energy, organic and inorganic. In taking the riddle to be solved — as OSTWALD²⁵ and others do — by the very setting up of the concept “nerve energy”, one merely introduces a

²⁴See more on this in my *Psychologi* II, 5; IV, 7 e; V B, 5; VII A, 1; B, 4–5.

²⁵OSTWALD seeks in his *Naturphilosophie* (Leipzig 1902) to show that the concept of energy is the fundamental natural-scientific concept, and since the phenomena of consciousness too must (with Kant) be conceived as expressions of activity (*Bethätigungen*), it becomes the fundamental psychological concept as well. But about the relation between the two kinds of energy he does not express himself clearly. Now the processes of consciousness are themselves called “energetic” (p. 394), now consciousness is said to be conditioned by the nervous system’s energy (p. 396), now spiritual energy is defined as “conscious or unconscious nerve energy” (p. 398).

qualitas occulta and acquiesces in it. The natural-scientific concept of energy is, wherever one meets it, always abstracted from phenomena with geometrical properties. Natural science knows energy only as an expression of the relation between phenomena in space. The riddle would therefore be solved only if we could form a concept of energy from which the psychological and the natural-scientific concepts of energy could be derived as special forms; but we lack the means to construct such a concept. In any case every attempt in that direction would carry us beyond the domain of the psychological problem.

II. The Problem of Knowledge

1. Not only psychological understanding, but all understanding, is conditioned by the relation between continuity and discontinuity. The latter conditions the fullness and the manifoldness of understanding's content; the former conditions its gathering-together and its ordering. Understanding appears under different principal forms, to which different kinds of science correspond.

I understand what something is when I recognize it again. Thus I understand, or know, who is coming when I recognize the one who is coming. Recognition rests on there being a connection between new and earlier experiences. In the act of recognition all intervening experiences are pushed aside, and the new phenomenon is identified — immediately or mediately, involuntarily or after some deliberation²⁶ — with a phenomenon that has occurred earlier. The contrast with what is new co-operates here as well; the quality of familiarity becomes the more strongly marked, the more unfamiliar the surroundings are. — It is recognition that makes *description* and *classification* possible; though the apprehension of difference co-operates essentially in both operations. In all describing and classifying certain types stand fast, at least provisionally fast, and the new phenomena are referred to them, either as entirely the same (coinciding with one another), or as similar (qualitatively alike), or as presenting analogy (likeness of relation). The positive concepts we form express such types. Where we can have recognition of no degree at all, and so can set up no type, we provisionally collect the phenomena under a negative concept — that is, we give them the common mark of being different from the types hitherto determined. The negative concepts have this significance in the history of classification, that by their help one gathers what has hitherto been undetermined into a group distinct from what is determined, so as then to leave it to later research to find positive determinations within it.²⁷

In the history of philosophy Plato's doctrine of Ideas stands as the characteristic expression of the significance of describing and concept-forming inquiry. The very

²⁶On the different kinds of recognition see my *Psykologiske Undersøgelser*, p. 35 f.

²⁷See *Det psykologiske Grundlag for logiske Domme*, p. 32 f.

possibility of recognition amid the motley manifoldness of the phenomena fills Plato with enthusiasm, and the highest spiritual capacity was for him that by which the different types underlying recognition are ordered according to their mutual likeness and difference, so that a world of thought is formed within which one can ascend and descend by steps that form a continuous series.

When concepts have been formed in different connections, they can be combined in judgments, and when different judgments have concepts in common that can be substituted for one another, inferences can be drawn. By the road of *inference* concepts and judgments can be formed without one's needing to go back to experience, as the describing and classifying sciences are obliged to do at every point. A continuity of a higher kind here asserts itself, in that thoughts and series of thoughts of the most various origin can be brought into connection with one another. While recognition rests merely on identity, inference rests on rationality: it is the relation between ground and consequent that governs, and that makes a new kind of understanding possible. I understand $A = C$ when I know $A = B$ and $B = C$. This kind of understanding is won through the formal sciences. It is peculiar to them that in the end it is a matter of indifference whence the first concepts and judgments derive, provided only they are of such a character that chains of inference can be built upon them. Just so, the describing sciences do not trouble themselves about where the phenomena come from, provided only they can be ordered according to relations of likeness and difference.

A third kind of understanding is characterized by this, that one neither merely ascends from phenomena to concepts, nor merely forms by inference new combinations of concepts out of given combinations of concepts, but derives new phenomena from phenomena given earlier. The new phenomena are understood by being set in a relation to earlier phenomena analogous to that which subsists between ground and consequent in an inference. This kind of understanding is characteristic of the exact real science (natural science), as it has developed especially since the age of the Renaissance. It is here the *concept of cause* that governs. We here connect experiences according to their unavoidable and law-determined succession in time.

This third kind of understanding has especially occupied the epistemology of more recent times, after HUME and KANT had impressed the difference between logical inference and real causal explanation. The possibility of forming and applying the concept of cause has in more recent times awakened a wonder similar to that which the possibility of forming general concepts awakened in Plato. It is, however, not only the intellectual need to find connection among experiences that has led to placing the concept of cause

first, but also the need to distinguish sharply between subjective representations and objective reality, since in doubtful cases the criterion of reality is always in the end the firm, unbreakable connection.²⁸ The world of dreams and of thoughts is larger than the world of reality (Eng ist die Welt, und das Gehirn ist weit), and men have felt ever more the need to get their thoughts so determined and so connected that they can stand as the expression of a reality — especially since only thereby can they hope to find the means of attaining their ends.

The concept of cause appears in two forms: a provisional, elementary form at which we are often compelled to stop, and an ideal form towards which all inquiry and all theories strive. *The elementary concept of cause* indicates merely an unavoidable relation of succession: when the phenomenon *A* has occurred, the phenomenon *B* thereupon unavoidably occurs, and *B* occurs only when *A* goes before. It is not thereby said that the causal relation obtains between *A* and *B* themselves; it is possible that both are successively appearing effects of an antecedent cause. *The ideal concept of cause* goes a step further and sees in the phenomenon called the effect a continuation of, or a conversion into new form of, the phenomenon called the cause. The ideal concept of cause thus passes over into the concept of development, wherefore it is no wonder that this concept, alongside the concept of cause, plays so great a part in the science of more recent times.

2. All three kinds of understanding rest on certain fundamental propositions or principles which involuntary thought-work — as it is practised by practical common sense, in the special sciences, in philosophical speculation and in religious thinking — feels no need to seek out, but which epistemology brings forward and tests. The problem of knowledge arises when it is asked what *the validity of understanding* rests on, and how far it extends. It is the relation between continuity and discontinuity that here again poses the problem. For Plato the problem arose from the irrational relation between concept and phenomenon; in more recent times, from the irrational relation between formal and real science. Can the concept ever be an adequate expression for the manifoldness of the phenomena, and the law for their changing combinations?

For PLATO the consequence was that the concepts (“the Ideas”) must have been known to man from a higher existence, and now emerged for thought in their whole perfection under the influence of the imperfect imitations of them which the world of sense presents. In more recent times the thought of nature’s lawfulness has often been taken for an a priori truth, an original intuition which at most owed to experience the

²⁸See *Det psyk. Grundlag*, Kap. 6.

occasion of its being called forth. It is the incommensurability between the principles (the logical principles and the principle of causality) and experience that has given birth to this speculative epistemology. This incommensurability has also given birth to a theory one might call the arbitrary theory, because it strongly stresses what is arbitrary in the first setting up of the principles — but which stresses no less strongly that the principles in their pure form can win no confirmation from experience, can never become results, but always only postulates. There can — says HOBBS, the most pronounced representative of this standpoint — be no science of the very principles of all science; they are set up by constructive art (*principia sunt artis sive constructionis, non autem scientiæ et demonstrationis*), and it is therefore we ourselves who bring forth their truth (*rationis prima principia vera esse facimus nosmet ipsi*). In FICHTE the first and unconditioned fundamental proposition of all human knowledge is set up by a free act; our science rests not on a “Thatsache” but on a “Thathandlung”, namely the thinking consciousness’s resolve always to remain in agreement with itself: the setting up of the principle of identity. Similarly S. KIERKEGAARD assumes a leap in the setting up of the principles, and KROMAN derives the principle of identity from the instinct of self-preservation, which seeks to maintain consciousness’s unity; “from experience it has in no way arisen”.²⁹

Both the speculative and the arbitrary theory must, however, concede that there must be determinate empirical occasions which call forth the intuitions or the postulates. Empiricism (which — when it takes account not merely of the single individual’s experiences but of the whole race’s — passes over into evolutionism) now lays, where it unfolds in its absolute form, the chief weight on these “occasions” and regards them as the complete causes. It is most clearly presented by STUART MILL and HERBERT SPENCER. It seeks to show how general principles may have formed themselves successively in consciousness under the constant influence of the surroundings, and it energetically maintains that no principle can have any validity beyond the confirmation by experience (verification) that it can win. Only as results, then, have the principles value.

What empiricism cannot explain is the fact that the principles themselves call forth experiences, through the questions they lead us to ask. More happily placed here is a fourth theory, developed in the most recent period by ERNST MACH and RICHARD AVENARIUS, which one might call the economic. It lets both passive experience and

²⁹HOBBS: *Logica*. Cap. 3 (§§ 8–9); *Physica*. Cap. 25 (§ 1). (Elsewhere, however, Hobbes teaches that it is analysis proceeding from what is given that leads us to the principles: *Analytica est ars ratiocinandi a supposito ad principia, id est ad propositiones primas vel ex primis demonstratas, qvot sufficiunt ad suppositi veritatem vel falsitatem demonstrandam*. *De rationibus motuum et magnitudinum*. Cap. 20 (§ 6).) — FICHTE: *Grundlage der gesamten Wissenschaftslehre*. § 1. — S. KIERKEGAARD: *Uvidenskabelig Efterskrift*, p. 83. — KROMAN: *Vor Naturerkendelse*, p. 270–274.

active development of thought come into their right, in that it regards the principles as those “conceptual reactions” which by the shortest road make intuitive apprehension and survey possible. When the principle of continuity plays so great a part, it is because it is a decidedly economic principle. There is here, then, only a question of practical applicability. One forms the concepts and principles one needs in order to make oneself master of the phenomena. Every practical and intellectual need is satisfied when our thoughts are fully able to reproduce the sensible facts. This reproduction is physics’s end and aim, and atoms, forces and laws are only means of easing the reproduction. Their value reaches only so far as they help us.³⁰ MAXWELL and HERTZ have expressed themselves in a similar direction.³¹

The four theories, which respectively conceive the principles of knowledge as intuitions, postulates, generalizations and economic instruments of thought, all presuppose *the analytic or regressive epistemology* which KANT in particular has developed. For *what* is to be the object of intuition, or of postulate, or of generalization, and *what* answers to the economic needs of research, can be discovered only by *inferring back from the data which experience presents, and finding the presuppositions on which the understanding of them rests*. Such an analysis must lie at the ground of every epistemology. Whether this analysis has been completed, one will not be able to convince oneself with full certainty. The history of science shows that here is a labour that must be taken up again and again. Now one supposes one needs more, now fewer principles than hitherto. A guarantee that one has reached the absolutely last presuppositions one will never be able to obtain. What the economic theory — to which I shall chiefly hold — particularly stresses is, partly that no more principles are to be set up than what is given demands with strict necessity (this is what is meant by “critique of pure experience”), partly that at different times, in different scientific situations, different principles may be needed, so that even a principle which has for a long time carried research forward may later be recognized as invalid, without its historical significance being thereby misjudged. The economic theory thus stresses partly the law of parsimony, partly the purposive, utility-determined character of the principles. It is due especially to two classes of motive: first, the wish to reduce to the least possible those presuppositions that cannot themselves be proved — an antidogmatic or antimetaphysical motive; and secondly, experiences from the history of the sciences, which show how principles and hypotheses may be legitimate and fruitful

³⁰ERNST MACH: *Beiträge zur Analyse der Empfindungen*. p. 144.

³¹MAXWELL: *Scientific Papers*. II, p. 360. — H. HERTZ: *Einleitung*. (Werke III, p. 1 f.). — Compare an interesting discussion *De la valeur des lois physiques* in *Bulletin de la société française de philosophie*. Mai 1901.

in a certain period but must later be replaced by others — something to which the most recent period's discussion of the grounding and the validity of the mechanical view of nature has particularly drawn attention.

There are, however, two sides of the matter which the special form the analytic epistemology has assumed in appearing as economic theory has not sufficiently stressed, and which become of particular importance when one does not regard the problem of knowledge in isolation but sees it in its connection with the other philosophical problems.

In the first place, the forms in which the intellectual need is satisfied must hang together with the general nature of consciousness. *What* we understand, and *that* we understand anything at all, depends not merely on the constitution of the phenomena but on our intellectual organization, just as which colours we see depends on the constitution of our organ of sight and not merely on the outward objects. There will be a certain type for all principles and hypotheses, one which points back in the end to the innermost nature of conscious life. And it will here always be the need for unity and continuity that one comes back to. Which and how many fundamental concepts (categories) and fundamental presuppositions one must set up is a problem that must always be taken up anew in knowledge's struggle to press forward. It was an illusion when Kant supposed that one could give, once for all, a catalogue of what is needed in this respect; but the old master was not mistaken in this, that it is *the need for unity and continuity* that lies at the ground of all the forms in which understanding is won, or is supposed to have been won. He has himself shown that all his categories can be traced back to the concept of continuity, and through all vicissitudes in the domain of principles it will undoubtedly be this concept that constantly recurs. The logical principles, the principle of causality and the fundamental propositions of natural science all lead back to this concept, which hangs so closely together with the nature of conscious life. A purely psychological epistemology will, however, never be able to suffice; for from the fact that the need for unity and continuity — be it ever so essential to conscious life — is satisfied by certain principles, it by no means follows that these are objectively valid. That need can find vent and satisfaction in many ways and in many forms, as the history of mythologies and of speculations sufficiently establishes — ways and forms which as a rule by no means satisfy the demands of economy, whether in respect of parsimony or in respect of applicability. Here SCHILLER's words hold: Eng ist die Welt, und das Gehirn ist weit. Of all the wealth of thought at the human spirit's disposal, only a narrow and firmly closed series of thoughts can be used when it is a question of valid understanding. What is a necessary presupposition is only this, that the presuppositions which the understanding

of what is given requires must be psychologically possible, must accord with the general laws of conscious life, and be only more special and more exact developments of what lies in those laws. A selection must take place among the thoughts that involuntarily press forward; but it brings with it no emancipation from the general psychological laws. What matters is that new dispositions of a special kind should form themselves — that there should arise an intellectual habitus which poses the questions and criticizes the answers in a stricter and more determinate way than the involuntary course of thought feels any need to do. Every comprehensive principle is really — psychologically regarded — the expression of such a habitus, which may be more or less deeply grounded in the nature of consciousness, that is, may now have the character of an instinct, now be due to the influence of practice. The purely logical principles come closest to the instinctive. The need for agreement with oneself, for consistency in the course of representations, cannot be explained from parsimony and expediency alone. Where strict inference from the experiences so far available leads us to self-contradictory results, we far sooner assume that the experiences are incomplete than that existence should contradict itself. What holds in the highest degree of the purely logical principles holds of more special principles as well. Thus the principle of causality expresses a tendency, when an event has occurred, to look about for other events in which the conditions of its occurrence might be found. Here too a need asserts itself to bring unity and connection into the content of consciousness. The more special and determinate the satisfaction of this need is to be, the more the economic principle operates in its two forms: as parsimony, which takes the straight road to the goal, and as expediency, which follows roads that really lead to understanding. Examples of this are to be had in Kepler's and Newton's demand for *veræ causæ*, in the principle of inertia, in the principle of energy, and so on. What purely empirically stands as a hypothesis becomes, epistemologically regarded, a principle, a guiding thought under whose leadership consciousness can, in the world of experience, satisfy its need for unity and continuity.

In the second place, principles and fundamental hypotheses — even if in the end they all stand as *working* hypotheses in the service of intellectual economy — must not be conceived as wholly accidental or arbitrary in relation to the existence which we mean to understand by their help. The very concept of a working hypothesis points in two directions: partly — as already shown — back to the nature of the thinking consciousness, since our consciousness cannot carry out a function, be it ever so economical, that is wholly foreign to its own essence; and partly outward, towards the existence to which the phenomena that are to be understood belong. A working tool must fit both the hand that

is to use it and the object that is to be worked upon. Existence must then in itself present sides that answer to the unavoidable presuppositions of our knowledge, however much these are conditioned by the inner and outer circumstances under which knowledge works — or perhaps precisely for that reason. This holds for all valid knowledge, from the most elementary forms to the highest. It holds already of the sense-qualities, despite their “subjectivity”, and it holds of the highest principles of unity in our thinking. KANT says of the tendency to presuppose a unity behind the differences of natural phenomena:³² “One might perhaps believe that we had here a merely economical contrivance of reason, serving to spare effort But such an egoistic purpose is very easily distinguished from the Idea according to which everyone presupposes that this unity of reason accords with nature itself, and that reason here does not beg but commands, although it cannot determine the limits of this unity.” I add, however: even where reason commands (or asks, expects, anticipates, postulates), it is obliged — as all commanders are — to adjust its commands to the capacity for obedience that may be presumed to be present. —

This last consideration leads in particular to making possible the connection between the problem of knowledge on the one side and the problem of existence on the other. It is all the more important to bring this out, since the analytic epistemology — like the economic, and in its way the arbitrary as well — leads to putting a new concept of truth in place of the current, naive one. The significance of the principles is to guide our work of winning understanding. Their *truth* consists in their *validity*, and their validity in their *working value*. That a principle is true means that one can work with it; and that means, when the principles of knowledge are in question, that by its help one can get forward in understanding — in firm ordering and connection of the phenomena. The concept of truth is a *dynamic* concept, in that it expresses a determinate way in which the energy of thought is employed, and a *symbolic* concept, in that it does not mean coincidence with, or qualitative likeness to, an absolute object, but likeness of relation (analogy) between the events in existence and human thoughts. The old naive concept of truth, according to which a piece of knowledge was true when it directly pictured or mirrored “reality”, is untenable, and has been so ever since the moment when men began to assert *the subjectivity of the sense-qualities*. That the sense-qualities were subjective was not, after all, to mean that they were invalid, unusable for orienting us in the world. They still stand as signs, signals, symbols, whose mutual succession we can interpret as the expression of an objective series of events, although they cannot be proved to be pictures of it. It stands similarly with the logical principles and the other

³²Kritik der reinen Vernunft. p. 653.

fundamental presuppositions of our knowledge. *Critical philosophy* led to the result — a simple consequence of the very analytic method it employed — that the truth of the fundamental propositions can mean only that they make scientific experience possible; they were themselves, after all, found by analysis as the necessary presuppositions of such experience. A comparison between our thoughts and an absolute world of things is not possible; we can compare only thoughts and experiences with one another. This consequence Kant himself did not see so clearly as some of his disciples did (Maimon and Fries);³³ the master himself was still entangled in dogmatism, as is seen from his doctrine of the “Ding an sich” as absolutely subsisting outside every subject. But when he designates the law of causality an “analogy of experience”, meaning thereby that events in time stand to one another as ground and consequent do in our thinking, he comes close to making it a working hypothesis. In the most recent period one is — as we shall soon see — *on the side of natural science* more inclined to acknowledge the dynamic and symbolic concept of truth than one was so long as the mechanical view of nature stood with a certain dogmatic character. — This new concept of truth, which is pressing forward in the domain of science, presents a kinship with the way in which — as we shall see under the fourth principal problem — the religious consciousness more and more places itself over against the dogmas; in the religious domain too it is more and more applicability and working value that are asked after. The static concept of truth³⁴ is everywhere being pushed aside by the dynamic.

Even supposing that fruitful principles or working hypotheses have been won, will existence then be exhaustively reproducible by their help, or will there always continue to subsist *an irrational relation between the principles our consciousness is able to form and existence itself, from which our experiences derive?* — It will appear that under three forms especially something irrational will always remain: in the relation between quality and quantity, in the significance the time-relation has for the concept of cause, and in the relation between subject and object. We shall consider these three points each by itself.

3. In the attempts to reduce all given differences to identity and continuity, the endeavour to trace differences of kind back to differences of degree has been of particular importance. In the science of material nature it expresses itself as an endeavour to explain all change as motion in space. Motion from one place to another is the simplest change; it would therefore be a great advance in clarity if it could be shown to be the change which, under different forms, takes place everywhere in nature where immediate experience

³³*Den nyere Filosofis Historie* II, p. 114 f.; 222.

³⁴The expression “static concept of truth” was used — with a somewhat different motivation — by LOUIS WEBER at the philosophical congress in Paris in 1900.

shows us qualitative changes. ARISTOTLE already taught that spatial motion lay at the ground of all other change, and of all coming-to-be and passing-away as well; he did not, however, hold it possible — as the Atomists had wished — to trace everything in material nature back to it. Only with the rise of modern natural science does this thought come forward in earnest. GALILEI stands as the great founder of what one is accustomed (in the narrowest sense) to call *the mechanical view of nature*. Besides the simplicity thereby attained — and Galilei was certain that nature follows the simplest ways — there is gained the further advantage that one can operate with determinate quantities. Galilei therefore said: measure all that is measurable, and make measurable what is not! By this reduction agreement becomes possible, whereas qualitative differences can only be estimated and not measured; and exact verification becomes possible.³⁵

The principles on which the mechanical view of nature rests were regarded by its most important representatives as absolute truths, fundamental laws of existence; in so far as it was still thought that they needed a grounding, this was taken directly from God's essence and will. In this Cartesians and Newtonians were agreed. And the Materialists differed from them only in holding the theological grounding superfluous and impossible. Through a series of magnificent discoveries and explanations this view of nature has shown its fruitfulness. The question for us is whether, from an epistemological point of view, it can be said to mark an absolute conclusion.

Even supposing that everything in material nature could be explained by the principles of the mechanical view of nature — and there is, as we shall soon see, doubt of that in the most recent period within the circle of natural scientists — it is in the first place clear that the qualities are not got out of the world by being “reduced” to quantities or charged to the account of the sensing subject. They remain standing as immediate facts which must be established empirically. The chemical product's properties cannot be derived from the properties of the elements; and when one kind of physical energy receives its equivalent in another kind of physical energy, the equivalent nevertheless has new properties which cannot be derived from the properties of the first form of energy. One cannot, however, let the sensing subject create these qualities out of nothing; in any case we then get insoluble psychological difficulties in place of the physical and chemical difficulties we have pushed aside.

In the second place, extension and motion are in the end themselves qualities in the widest sense — factual properties which in and for themselves might just as well need an explanation as the specially so-called sense-qualities. Since BERKELEY's and

³⁵*Den nyere Filosofis Historie* I, p. 160–167.

LEIBNIZ's time this has often been urged from the philosophical side. There is no reason to assume that the quantitative properties express the innermost essence of things. The reason science seeks them out by preference and holds to them is really only that by their help one can give exact descriptions of the material phenomena and draw determinate inferences from them. This has in the most recent period been strongly emphasized by investigators such as MAXWELL and HERTZ — in the latter's case with express adherence to the economic epistemology discussed above. "The progress of the exact sciences", says Maxwell, "depends on the discovery and development of appropriate and exact ideas, by whose help we can form a mental representation of the facts which is at once sufficiently comprehensive to stand for every particular case, and sufficiently exact to afford security for the inferences we draw from them by way of mathematical calculation." And Hertz says: "We form for ourselves such inner images (*Scheinbilder oder Symbole*) of outward objects that the logically necessary consequences of the images are always in turn images of the naturally necessary consequences of the objects imaged Any other agreement with the things, the purpose of our representations does not require."³⁶ — The dynamic and symbolic concept of truth is here expressly put in the place of the naively dogmatic one which the mechanical view of nature had formerly espoused. What matters is only to find a circle of symbols that can be applied with full consistency, and from which inferences can be drawn that are confirmed by new experiences, which in their turn can themselves be expressed by the same circle of symbols. But the possibility can here never be ruled out that another circle of symbols would be able to express the positive experiences equally well or better, and to serve as the ground of exact and verifiable deductions. It will never be possible to prove of any circle of symbols that it is the only right one, and the only necessary or possible one.

The epistemological reflections to which recent natural scientists have thus been led were called forth by the difficulty that presented itself in bringing the electrical phenomena under the mechanical view of nature, or else in deriving that view's principles from the laws holding for those phenomena. HERTZ inclines rather to the latter possibility, BOLTZMANN rather to the former.³⁷ But of greatest interest for epistemology is MAXWELL's criticism of the current concept of matter, according to which by matter one imagines an extended mass. The fault in this concept is, according to Maxwell, that it leads one

³⁶See Note 31.

³⁷H. HERTZ: *Ueber die Beziehungen zwischen Licht und Elektrizität*. 1889, p. 29. — BOLTZMANN's lecture at the German congress of natural scientists 1900 (translated in *The Monist*, January 1901: *The recent development of method in theoretical Physics*) and his paper in the Vienna Academy's *Oversigter* CV, 8 (translated in *The Monist*, Octbr. 1901: *On the necessity of atomic theories in Physics*). — On the whole question compare also C. CHRISTIANSEN: *Den elektromagnetiske Lysteori*. (Det danske Vid. Selsk.s Oversigter. 1889), and the same author in *Fysisk Tidsskrift* I, p. 4–5.

to think of matter as something at rest. But it is motion that makes rest intelligible to us, not the reverse. The doctrine of motion must therefore precede the doctrine of equilibrium, dynamics precede statics. It is from motion that we win the concept of force or energy, by whose help equilibrium becomes intelligible to us. If now by matter we understand what is constant, what subsists through all changes, then it can only be the law of motion itself, and the essence of “matter” consists accordingly in motion. It is, according to Maxwell, a prejudice to regard matter as extended and the molecules as hard and solid: one would then have to ask what holds the molecule’s parts together, and one would arrive at molecules of a second order. Yet we must constantly operate both with geometrical and with dynamic concepts, whether we regard the last elements as extended or not. It is only extension as a resting property that Maxwell declares against; even with a line we draw on a board, the motion by which it comes to be is the essential thing.³⁸ — It is an extraordinarily important epistemological point of view that is here brought forward: that resting states always contain problems which can be solved only by rest’s being replaced by motion. Potential energy we understand only through actual energy, capacities and dispositions only through what comes of them. This law holds everywhere, in the psychical domain as in the physical. There is a half mystical, half materialistic inclination to find completion in contemplation of — or staring at — a resting “something”.³⁹ Maxwell has made an important contribution towards getting this inclination rooted out. But the great question is whether the thought of the continuity of motion, or of activity, can be carried through in all domains. Even if the dynamic asserts, epistemologically regarded, its priority over the static, the static cannot for all that be got rid of; it presents itself ever anew with its tasks for thought, and Maxwell himself acknowledges it when he declares both geometrical and dynamic concepts necessary in the explanation of nature. In contrast to the dynamic, the geometrical designates simultaneity. In the domain of material nature the simultaneous appears in the form of space;⁴⁰ in the psychical domain the relation of simultaneity does not assume this

³⁸MAXWELL: *Scientific Papers*. II, p. 26–33; 302–305 (where M. shows what he owes to Faraday and Lord Kelvin); 326; 777 f.

³⁹Compare my *Psykologiske Undersøgelser*, p. 77–81.

⁴⁰This point OSTWALD does not let come into its own when he wants to trace everything in nature back to “energy”. Mass, the filling of space, weight and chemical properties are all supposed to be traceable back to the concept of energy. It is correct that all those concepts presuppose the concept of energy; but is that the only fundamental concept in natural philosophy? In that case one would have to be able to derive the “geometrical” properties from the dynamic ones, and at that Ostwald makes no attempt. In his work *Die Ueberwindung des wissenschaftlichen Materialismus* (1895) he even defines matter as “a spatially [sic] ordered group of different energies” (p. 28). This spatial ordering, here acknowledged as a special moment, is — it seems to me — somewhat pushed aside in his *Naturphilosophie* (1902), where “the energetic world-picture” is developed. Yet his main proposition, on which he builds his system, is that “every process without exception can be exactly and exhaustively described or presented by indicating

geometrical form, but appears nonetheless as something “static” which sets research new tasks when it has happily and successfully found the laws of the psychical processes. After the continuity in the processes of development has been exhibited, it remains to exhibit that there is continuity between the processes and the resting states. — We meet here again, then, the standing problem; it would present itself even if the qualities were fully explained by being represented by quantities, and even if the fundamental physical concepts with which science operates were more than the most perfect and most expedient circle of symbols that it has so far succeeded in forming and applying. The possibility of an irrational relation between existence and our knowledge cannot therefore be ruled out.

4. The investigation of the relation between *the elementary and the ideal concept of cause* will lead us to a similar result. In its elementary form the concept of cause indicates, as we have seen, only this relation between two events: that when the one has occurred, the other unavoidably occurs. But from this unavoidable succession research advances, through observations, experiments and hypotheses, so far as possible to the exhibition of a continuity in the series of events, by which the differences between the single members of that series become vanishing, so that at last even the temporal difference between the events is reduced to the least possible. From outer, though unavoidable, succession thought works its way through continuity towards a complete identity. While the time-relation plays an essential role in the elementary concept of cause, it is as good as eliminated in the ideal concept of cause. If this process could be carried through completely, we should arrive at the paradoxical result that the perfect causal explanation would be an abolition of the concept of cause itself: for the causal relation differs from the purely logical relation between ground and consequent only by the temporal difference between the members of the relation.

The philosophy of the seventeenth century (SPINOZA and LEIBNIZ) did not distinguish between cause (*causa*) and ground (*ratio*); the causal relation between two phenomena was for them the same as the relation between premisses and conclusion in an inference, a mere relation of identity. It stood for them as self-evident that there could not be more in the effect than there was in the cause; the time-relation between the members was

which energies undergo temporal or spatial [sic] changes” (p. 152). That the concepts energy and time cannot be separated is a matter of course, since energy means the capacity to overcome a resistance, and every overcoming takes time. Not so much a matter of course is the connection between the concept of energy and spatial changes. This connection derives, in Ostwald, simply from his having abstracted the concept of energy from spatial phenomena, from investigations of changes at determinate places in space. Only by this geometrical moment’s being pushed aside in his concept of energy does it become possible for him to believe that the concept is without further ado the same for psychical and for physical phenomena, so that the problem of “soul and body” would be solved by setting it up. See above, Note 25.

indifferent, and time on the whole belonged only to the obscure sphere of sensuous representation (imagination). When HUME raised the problem of causality, he did so precisely by emphasizing that no proof can be given for the justification of inferring from past to future. When appeal is made to the experiences of the past, Hume says, these can only show that a thing has at some time, in a single moment, been in possession of a certain power, not that it constantly possesses it and will possess it.⁴¹ KANT believed indeed that he could give a rational proof of the validity of the causal principle, but he distinguished between the causal relation and the relation between ground and consequent, in such a way that there was only an analogy, not an identity, between them: the law of causality meant for him that events follow one another in time in a manner analogous to the manner in which the conclusion springs from the premisses. The causal principle therefore gives us a rule by whose help we can attain unity in our experiences.⁴²

From two different standpoints attempts have been made in the most recent time to eliminate the time-relation and thereby to abolish not only the elementary concept of cause, but in the end the concept of cause altogether.

From a speculative standpoint it has been maintained that the time-relation always points to an imperfection, to an incompleteness in development. So long as the time-relation is determinative for our conception of existence, our conception is therefore always imperfect, and the reality of the time-relation is irreconcilable with the idea of a perfect knowledge, of an absolute truth. Our knowledge does in fact work itself more and more away from the time-relation; the more clearly we understand something, the smaller the role the temporal difference plays — the more does real knowledge pass over into formal knowledge, cause into ground. What we begin by calling cause and effect comes, as knowledge advances, to stand as members in a whole, members which stand in a lawlike relation to one another, a rational relation for which the temporal sequence is unessential. That we first observe the one member and thereafter the other has no significance: in our knowledge of the law the whole process stands before us as a unity. It is not the time-relation, but the unity behind the time-relation, that binds what

⁴¹HUME: *Treatise on human nature*. I, 3, 6: Your appeal to past experience decides nothing in the present case, and at the utmost can only prove, that that very object, which produc'd any other, was *at that very instant* endow'd with such a power.

⁴²*Kritik der reinen Vernunft*: p. 181: Wir werden also durch diese Grundsätze [the causal principle and the kindred principles concerning the persistence of "substance" and the reciprocal action between all that exists] die Erscheinungen nur *nach einer Analogie mit der logischen und allgemeinen Einheit der Begriffe* zusammen zu setzen berechtigt werden. — p. 180: Eine Analogie der Erfahrung wird also nur eine Regel sein, nach welcher aus Wahrnehmungen Einheit der Erfahrung entspringen soll, und als Grundsatz von den Gegenständen (der Erscheinungen) nicht constitutiv, sondern nur regulativ gelten.

we call cause to what we call effect. Besides, neither is there any interval between the end of the event we call cause and the beginning of the event we call effect. It is the English philosophers FRANCIS BRADLEY and BERNARD BOSANQUET⁴³ who have urged this consideration, at the base of which lies a speculative interest in a total conception and in concluding concepts.

From a wholly different standpoint, which may be designated the standpoint of “pure experience”, and whose most important representatives are RICHARD AVENARIUS and ERNST MACH, work is done in a similar direction, but from other motives. The interest here is directed to separating out all the accessory representations and auxiliary concepts with which we, involuntarily or on methodological grounds, supplement experience itself, the immediately given. The advance of knowledge consists in a reduction of the differences (to a “heterotic minimum”) and in an approximation to a pure description of a continuous process. In the course of this advance the content of knowledge is more and more restricted to descriptive statements, so far as possible with analytic transitions, and the differences are reduced from qualitative to quantitative, so far as possible with a constant exhibition of relations of equivalence. It is no longer the task of strict science to “explain”; whoever wants “explanations” is referred to mythology and metaphysics; the aim of science is to give an exact, methodical description of all relations and transitions. As regards the causal relation in particular, weight is laid partly on this, that every application of the concepts cause and effect arbitrarily draws two elements out of the whole connection in which they occur and sets them in opposition to one another; partly on this, that the time-relation may very well be reversed as soon as one has exhibited a relation of equivalence, and as soon as one abstracts from the direction of the changes.⁴⁴

The epistemological views which thus, with different motivations, would eliminate the elementary concept of cause and deny the fundamental significance of the time-relation may be characterized by this, that they put an ideal of knowledge in the place of the actual knowledge we can attain at any given time. Every determinate investigation must begin at a determinate point, which lies where the problem presents itself. And the problem is set when two members in the series of events draw attention to themselves and awaken surmises of an inner connection between them, or when a single member

⁴³FRANCIS BRADLEY: *Appearance and Reality*. London 1893. Chap. 4–7; 18. — BERNARD BOSANQUET: *Logic or the morphology of knowledge*. Oxford 1888. Book I. Chap. 6.

⁴⁴RICH. AVENARIUS: *Kritik der reinen Erfahrung*. II. p. 332–339. — ERNST MACH: *Zur Analyse der Empfindungen*. p. 168: “Die Thatsache der Nichtumkehrbarkeit der Zeit reducirt sich darauf, dass die Werthänderungen der physikalischen Grössen in einem *bestimmten Sinne* stattfinden. Von den beiden analytischen Möglichkeiten ist nur die eine wirklich. Ein metaphysisches Problem brauchen wir hierin nicht zu sehen.” — Compare on these theories HEINRICH GRÜNBAUM: *Zur Kritik der modernen Causalanschauungen* (Archiv für systematische Philosophie. 1899. p. 392–409).

steps forth in its peculiarity or its suddenness and thereby awakens the need to seek intermediate members through which it can be brought into connection with the whole remaining series. It is to that extent an accidental, arbitrary starting-point that is seized upon; but from the nature of the case it cannot be otherwise. Our knowledge develops historically, because human attention is awakened only under certain determinate conditions. If it were directed without difference and at all times equally upon all members in the series of events, no knowledge would arise. It is of course of importance that one be conscious of what is accidental or subjective in the starting-point and in the section taken; and if the work of knowledge has success, one does get beyond it, since the section is precisely through understanding worked into a great continuous connection.

The same historical character of our knowledge becomes clear when we bethink ourselves that we always stand between the experiences of the past and the possibilities of the future. *In each single moment* we must distinguish between what lies given and what we expect; the former is the cause, the latter the effect; and the expectation can never become more than a hypothesis. The single moment, in which on the one side a “no longer” holds and on the other side a “not yet”, sets the problem before us in its whole tension, a tension which only the power of habit diminishes. With new problems the tension will not fail to appear — and so long as knowledge goes forward, it will seek and find new problems.⁴⁵ The full connection is always present only afterwards, and until it is present the assumption of it stands as a hypothesis. We shall therefore never be able to dispense with concepts such as force, energy, cause or possibility, which with different nuances express one and the same relation, namely the dependence of later states on the preceding states; this dependence differs from purely logical and mathematical relations of dependence in that it is at once a time-relation and a rational relation, since the following state both follows after and follows from the preceding one. Even if the continuity be exhibited by the help of ever so many intermediate members and nuances, this relation continues to hold for every little transition between two members or nuances.

And by this nothing is altered, even if equivalence is exhibited between the states that succeed one another. Hume’s problem has by no means, as it has sometimes been said, been solved by the discovery of the conservation of energy. A relation of equivalence does not exclude a qualitative difference, but precisely presupposes it: I have no ground for setting up the judgment $A = B$ when A and B have not first stood before me as different, before I found on closer investigation that they could be put in the place of one

⁴⁵BOSANQUET accordingly concedes this too: *At any given moment* we have no choice but to say that the future is conditioned by the past, ... effect by cause. *Logic* I. p. 272.

another. When such a relation of identity has been found, it is indifferent whether we go from *A* to *B* or from *B* to *A*; but at the time when it was to be found, we began with *A* or *B* and came by way of investigation over to the other concept. In a judgment we must therefore distinguish between the psychological process by which the judgment comes to be, and in which a determinate difference asserts itself between the initial representation (subject) and the concluding representation (predicate), and the finished judgment, which can be formulated as a relation of identity in which the difference between subject and predicate becomes indifferent.⁴⁶ And just as little as in the movement of thought is the order of the members indifferent in outer happening. If I have recognized that there is a relation of equivalence between heat and motion, it is, purely abstractly regarded, indifferent whether I go from heat to motion or from motion to heat; but “the direction of the changes” is in reality a question of life and death: existence actually becomes another thing according as the changes proceed preponderantly in the one or in the other direction. Time then cannot in reality be reversed, and therefore the exhibition of relations of equivalence cannot form the conclusion of our knowledge. We must in addition have information about the actual direction of the changes (the conversions), and that is won only by ever new experience.

There comes in addition that every relation of equivalence — as well as every causal relation whatever — comes into force only when certain conditions, and particularly releasing conditions, are present. With respect to the presence of these conditions a new problem then arises: can it be shown, with respect to them too, that the relation to their cause may be presented as a relation of equivalence? We come here into an infinite series, where the answers (the determinate, concrete answers) cease long before the questions cease. Here too we have a purely logical analogy; for every absolute relation of identity ($A = B$) between two concepts rests, on closer consideration, on determinate conditions (so that it — according to JEVONS’s formulation — must be expressed by $AC = BC$);⁴⁷ it then becomes a new question how matters stand with these conditions (*C*), and so thought goes ever further. —

There is thus no prospect of escaping the historical element in our knowledge. The measure of the development of knowledge consists in the extent to which the elementary concept of cause (unavoidable succession) can be applied, as compared with the mere ascertaining of actual simultaneous or successive differences, and next in the extent to

⁴⁶See *Det psykologiske Grundlag for logiske Domme*. p. 46–50.

⁴⁷JEVONS: *Principles of Science*. 2. ed. London 1877. p. 43. (Already DE MORGAN, as Jevons remarks, has said that we usually think and argue within a limited world or sphere of concepts, even when this is not expressly declared.) — Compare my *Formel Logik* 3, § 16 and § 22.

which the elementary concept of cause can be replaced by the ideal concept of cause (equivalence or identity). But the process of knowledge will at every time consist in an ascent through the three stages here indicated — a process of ascent which must ever anew be repeated from new starting-points. And it is the reality of the time-relation that conditions this necessity. Speculative conceptions — whether they appear in idealistic or in realistic form — have therefore always an inclination to push the time-relation aside or to regard it as merely “empirical”, if not as illusory.⁴⁸ Should it be an illusion, then it is in any case an illusion in the second power when one believes one can free oneself from this illusion. For us existence never dissolves wholly into thought.

5. From yet a third side the problem of knowledge shows itself in its whole sharpness, at the same time as a continual development of knowledge shows itself possible. In every act of knowledge a distinction can be drawn between *a subjective and an objective element*, between that which knows and that which is known — but both elements are given only in relation to one another, though within this relation they can assert themselves in different degrees.

What now is subjective, and what is objective, in our knowledge? — Already from the foregoing it will be seen that this question is answered in different ways. In the domain of science a tendency asserts itself to charge all differences of quality, everything that breaks continuity, and at last perhaps even everything that breaks identity, to the subject’s account. The sense-qualities and the differences of space and time would then be merely subjective. What is justified in this way of looking at the matter rests on this, that the differences we posit in qualitative, extensive, intensive and protensive (temporal) respects depend on our psychical dispositions. Our sensation is a discriminating whose result depends on the organization and the previous history of the sensing subject. The differences found hold, therefore, only in relation to the standpoint of the knowing subject in those various respects. Add to this that our attention is discontinuous, moves by leaps to and from its object; by that too there come forward differences and interruptions which cannot be charged to the object’s account.

The dogmatic and speculative tendency in philosophy and in natural science has

⁴⁸FRANCIS BRADLEY has clearly seen the contradiction between time and “the Absolute” when he declares: If time is not unreal, I admit that our Absolute is a delusion (App. and Reality: p. 206). (The Absolute has no seasons. ib. p. 500). — The critique of, and the reaction against, a speculative tendency therefore also often supports itself on the reality of time as a main argument. Compare C. H. WEISSE’s and S. KIERKEGAARD’s relation to Hegel (*Den nyere Filosofis Historie*. II. p. 243; 261 f.) In his acute work *Tidsexistensens Apologi. Et stykke relationsteori* (Upsala 1888) PONTUS WIKNER has attempted to shake the either-or that here presents itself. He wants to show that the highest perfection can be attained only by a successive unfolding of qualities which as simultaneous would exclude one another. But the limitation that lies in succession itself is not thereby abolished!

been inclined to take this road. Emancipation from the merely subjective became here at the same time an approximation to continuity and identity as the essence and the norm of truth.

The critical philosophy emphasizes, in contrast to this, that the qualitative, extensive, intensive and protensive differences make up the material given for our knowledge and set us the tasks of our inquiry. Just as our personality works at gathering its sporadically appearing elements, at harmonizing the conflicting tendencies and freeing itself from what is obscure and self-contradictory, so our understanding works at transforming the given differences, which for us are matters of fact, into stages in one and the same continuous process of development, or into forms of one and the same content. The need for continuity and identity lies deep in human consciousness, and consciousness therefore seeks to find them again in the given content of knowledge. Moreover, consciousness cannot itself bring forth the differences which form the material of its working, however much the form and the degree in which they appear in consciousness are determined by consciousness's own involuntarily operating conditions. — In the discussions of the most recent time concerning the epistemological foundation of natural science there shows itself, too, a tendency to derive all points of view of unity preponderantly from the economy of the knowing subject.

In reality we never at any time have the pure subject with its forms given in opposition to a pure object, or rather to a “thing in itself” from which the manifoldness of the content of knowledge would derive. KANT concluded here too quickly, in supposing that the subjective forms of knowledge could be determined once for all, so that the “matter” in knowledge would have to be derived from “das Ding an sich”. In the more special development of his theory of knowledge he could not, however, avoid coming to ask whence the forms derived, and to hint that these too (which are, just as much as the matter, factually given, since they must after all be found out by psychological analysis) are ultimately determined by the “Ding an sich”; and since the “forms” designate what is constant in our knowledge, the last presupposition of Kant's philosophy becomes something for which he can really give no grounding, namely that the “Ding an sich” works in a constant manner, since otherwise the forms could not subsist, or in any case could not be applied.⁴⁹ Thus Kant comes to oscillate between the two views described above, though it was plainly his intention to hold to the latter. Continuity was for him the common fundamental form of the categories, and synthesis was for him the fundamental law of the working of the knowing consciousness.

⁴⁹See my treatise *Kontinuiteten i Kants filosofiske Udviklingsgang*, p. 26–35.

The problem is more involved than Kant saw. When we distinguish between subject and object in our knowledge, it is in reality an objectively determined subject (S_o) that we set in opposition to a subjectively determined object (O_s). The properties or “forms” we ascribe to the subject cannot be explained from the concept of subject itself (the pure S); they stand as facts, just as much as all the other properties our knowledge gets to deal with. And the properties or determinations we ascribe to the object belong to it always only in relation to a subject, and, more precisely, to a subject of a certain special constitution. Therefore the problem begins ever anew: whence does the subject get its objective (factual) properties, and what relation is there between the object’s subjective determinations (qualities and so forth) and its essence, as this would come forward before subjects of another kind than we?

Here again we meet the irrational, and here we see perhaps most clearly how inexhaustible existence is in relation to our knowledge. What was justified in Kant’s setting up of the concept “Ding an sich” lay in this, that a limit-concept is needed to express the irrational relation between what he called the “form” of our knowledge and its “matter”. If we would keep the concept “thing in itself”, we can employ it in Kant’s spirit and yet avoid falling into the contradictions with which it is burdened in his philosophy, if we use it to express the fact that the difference between subject and object always comes into force anew, however often we may have found an objective explanation of the subject’s peculiarities (of S_1 by O_1) and a subjective explanation of the object’s peculiarities (of O_1 by S_2). The irrational shows itself in this, that a continual series-formation (of the type: $S_1 \{ O_1 \{ S_2 \{ O_2 \dots \}$) is possible and necessary. Thought must ever anew set about finding predicates for the determination of existence, because the source that makes the stream of thought possible is inexhaustible. “The thing in itself” is the obscure, tacitly understood initial representation, which comes forward ever in new ways and demands new determinations.⁵⁰ Perhaps the right symbol for the relation of our knowledge to existence would not be an irrational number, but an imaginary number, since existence might have attributes which cannot at all be reached or determined in the dimensions in which our thought is able to move. The possibility of this cannot in any case be disproved. Nor can that other possibility, that existence is rational only in a very slight degree, and that it might one day turn towards us another side, one that would not make possible the building of coherent and applicable series of thoughts; there would then, as I have expressed it in another place,⁵¹ begin a logical ice age for us. The relation between

⁵⁰Compare *Det psykologiske Grundlag for logiske Domme*. p. 39 f.; 46 f., and (as regards religious judgments) *Religionsfilosofi*, p. 67–71.

⁵¹See my treatise *Filosofien og Livet*. (Tilskueren 1901). p. 34. — On the imaginary in an epistemological

subject and object would not come into force at all; neither an S_o nor an O_s could subsist.

When in an earlier connection I used Schiller's words: Eng ist die Welt, und das Gehirn ist weit, it now shows itself that the sentence can be reversed: Gross ist die Welt, und das Gehirn ist klein! Knowledge is, however rich and powerful it may be, always only a part of existence, and the problem of knowledge could be solved only if existence as a whole (in so far as it can be thought as a completed whole!) could be expressed by the help of a single part of it. Symbolic the expression must in any case always remain. Our knowledge gives us, even where it reaches highest, only a section of something more comprehensive. Of all the possibilities of thought only a single one is represented in the reality we know; but the reality we know is itself always only a part of a greater whole — and the relation between part and whole we cannot here determine. We can form no exhaustive concept of reality.

sense compare WUNDT: *System der Philosophie*. p. 195–199.

III. The Problem of Existence

1. Both the problem of consciousness and the problem of knowledge point out beyond themselves: the life of consciousness is a part of existence, and it is the task of knowledge to understand existence. It is then natural to ask what place the life of consciousness occupies in existence, and what picture knowledge is able to give us of existence. The problem to which we are thus led over we may call the *cosmological*, since it is the problem whether a completed world-view is possible: *Kosmos* means existence regarded as a whole, and *Logos* means doctrine, view. The word metaphysics it will also be warranted to use here, though it has been employed in many different senses and is on that account less clear.

The chief difficulty in this problem will be that the cosmological principles, the principles which are to lie at the base of the world-view, cannot without more ado be fetched from any special domain of experience, since the point would precisely be to connect all domains of experience into a whole and to give a positive characterization of it.

An attempt to treat the problem of existence will have both a formal and a real character: partly it must be asked what demands must be made on the concept that is to be able to embrace existence as a whole, and whether it is possible to satisfy those demands; partly it must be asked what positive marks can be assigned of such a whole, if we are able to form a concept of it.

The *formal* motives for speculation on the nature of existence lie deep in the constitution of consciousness and of knowledge. They hang together with the need for continuity, the need in which personality and science meet. The very essence of thinking expresses itself on all levels and in all forms as a gathering-together, a synthesis, and it is then not to be expected that thinking should of its own accord halt in its attempt to work together what is sporadically given. This striving acquires a special and practical significance from the fact that a firm and continuous connection is the only criterion we have in doubtful cases when it is a question of distinguishing between dream or illusion and reality. The more comprehensive and the more inwardly coherent a concept

of reality we can form, the greater the certainty it contains. There will therefore, on theoretical as well as practical grounds, be an inclination to go to the limit and to attempt to continue and to complete what theoretical and practical orientation already leads us into. The ideal would be reached if a complete harmony could be exhibited between all our experiences, a continuous whole into which all the special domains of experience join themselves according to their own laws.

But the discussion of the problem of knowledge must already have taught us that such a completed world-view is impossible, and that in a certain sense it contains a self-contradiction. None of the special domains of experience lies before us completed; new experiences and new riddles are always coming up; our comprehending thought is always getting new tasks. And since our knowledge always works by holding things together and comparing them, every total picture, if it were to be an object of completed knowledge, would have to be held together and set beside something that was different from it: only so could it get its full determinateness; but if there were anything different from it, it would after all not be a *total* picture! — It is here indifferent whether we stop at the totality we have attempted to form, or whether we go back to the principle of such a totality and call it “God” in contrast to the totality itself as “the world”; the antinomy remains the same in both cases.⁵² The irrational announces itself here again, as it did at the problem of knowledge. And there is here surely an inner connection. That knowledge could not be completed might hang together with existence itself not being completed, not being finished, but being in continual becoming, as the single personality is, and as knowledge is. Perhaps it also conceals simultaneous disharmonies within itself, which make it impossible for it to constitute a harmonious whole. The analogy between the different problems would in that case be especially clear. In the various systems men have been too sure that existence in itself was a closed and constant whole, and that it was only our will and our thought that had always to struggle in order to subsist and to win harmony.

A *real* motive for speculation lies in the prominent role which a phenomenon, or a domain of experience, or a side of existence can play for us. From this there can arise a tendency to lay this phenomenon or this side at the base of an interpretation of existence, and to attempt to derive the other phenomena or sides from it, or in any case to trace them back to it. The different cosmological systems are so many attempts to sound the depths of existence, to test how far a thought can cast light over existence. They are a series of thought-experiments by which the bearing power of our most important

⁵²Compare on this my *Religionsfilosofi*, p. 57–65. — In the epistemological section of the *Religionsfilosofi* I have on the whole already set forth my conception of the problem of existence and of its treatment.

thoughts, their capacity to serve as the foundation of a comprehensive world-view, has been tried.

Every such attempt goes — as LEIBNIZ first saw with brilliant clarity — necessarily by the way of *analogy*. — In all science analogy can become of importance. All designations for psychical phenomena were originally formed on the basis of an analogy with physical phenomena, and in psychology we are constantly compelled to operate with physical analogies; it then becomes an important question how far their validity reaches. The causal principle expresses, according to Kant, that between the actual events in the world there is a relation analogous to that between ground and consequent in our thinking. The atomic theory, and on the whole the so-called mechanical view of nature, is in the most recent time regarded epistemologically as a great analogy, by whose help the qualitative changes in nature can be described and calculated. In discoveries too analogy plays a great role: the analogy between chemistry and physics helped Robert Mayer to his discovery of the conservation of energy. But when analogy is employed metaphysically or cosmologically, it is not *one particular* domain of existence that serves to illuminate *another particular one*; it is *one particular* domain that is used to express existence *as a whole*. This symbolizing becomes of another kind and another validity than the symbolizing which finds application within the special domains: it cannot be carried through with full consistency, and it cannot be verified. Neither the warrant of the analogy nor its limitation can be strictly exhibited. In this the cosmological or metaphysical symbols differ from the scientific ones. — As I have sought to show in my *Religionsphilosophi*, the religious symbols share the fate of the metaphysical. In both cases an attempt is made to form concluding concepts with real validity; the difference rests only on the motive that leads to these attempts.

In the problem of knowledge we came likewise to an irrational relation between part and whole. There it was a question *how far the whole could be expressed in and for a single part*. Here, at the cosmological problem, it is a question *how far determinations can be fetched from a single part which hold for the whole as such*. It is one and the same relation, coming forward in the two problems from different sides. —

The character and the value of a world-view depend not only on how clearly and consistently the analogy is carried through, but also on where the analogy is fetched from. The phenomenon (the part or side of experience) on which the analogy is built we may call the *primal phenomenon*. This expression stems from GOETHE. But Goethe understood by a primal phenomenon not merely a phenomenon of a typical kind, able to illuminate other phenomena for us; it means for him a fact which is at the same time

a law, so that one need only utter it in order to have explained it, and it must not be regarded as composite, since it is itself to be the symbol of everything else, to be a “limit of our beholding” which at once awakens “the great wonder” and the feeling that we stand at the limit of our capacity!⁵³ — It is well known what a fateful role this concept played in Goethe’s theory of colours. By his very definition of it he meant to exclude every further investigation and explanation of the phenomenon he chose as “primal phenomenon”. But this defect need not cleave to it. It may very well designate the point in our experience from which we attempt to orient ourselves in all directions, and yet be itself an object of further investigation and explanation. The main thing is that the phenomenon in question should be so peculiar and so significant that it can take on a typical character for us. The interpretation of existence must always proceed from a single domain, and it can proceed without that domain being itself withdrawn from special scientific treatment.

But where now are we to seek the primal phenomenon? On that turns the battle between the different world-views. Now it is life, now thought, now matter that is made the primal phenomenon and laid at the base of the interpretation. Before we go over to considering the most important primal phenomena that have been taken into the service of cosmological interpretation, we must nevertheless dwell a little on this interpretation itself.

2. Metaphysics can become dogmatic, and thereby work harm in a scientific respect, in two ways — partly by cutting off every special scientific explanation of the chosen “primal phenomenon”, partly by conceiving its analogizing method as more scientific than it is, and forgetting the necessity of continual confirmation by experience. It cannot interfere in the housekeeping of the special sciences, in that of the human sciences as little as in that of the natural sciences. Its philosophical place is at the boundary of scientific thinking; its task is to give a last interpretation. The theory of knowledge led us to a limit-concept for that which conditions the continual conflict in our conception of the world between quality and quantity, the elementary and the ideal concept of cause, subject and object. If we choose to call that which contains the ground of these oppositions “the thing in itself”, then this expression designates the philosophical place of metaphysics or cosmology.

The last interpretation will have more of the stamp of art than of science. The

⁵³GOETHE: *Farbenlehre*. § 124; 175–177. Compare *Gespräche mit Eckermann*. 18. Febr. 1829; 21. Decbr. 1831. — Goethe applied this concept not only in the domain of the theory of colours, but in magnetism and in botany as well. See *Sprüche in Prosa*. (Ueber Naturwissenschaft) and *Gespr. mit Eckermann*. 27. Januar 1830. — HEGEL was (as is seen from Goethe’s *Nachträge zur Farbenlehre*) much taken with the concept of the primal phenomenon as Goethe had set it up.

transition from the empirical and critical parts of philosophy (psychology and the theory of knowledge) to cosmology presents a certain analogy to the transition from historical criticism to the writing of history. The historian forms the critically treated material into a total picture of events and characters; what is fragmentary is filled out, and a rounding-off is given in which the personality of the one who presents it necessarily plays its part. The cosmological philosopher will in the same way give us a total picture of existence on the basis of our knowledge of it; he will work the scattered features together into a totality within which a single element (the primal phenomenon) is particularly determinative, so that the timbre of the whole depends on it. In the choice of the primal phenomenon and in the carrying through of the analogy a decidedly personal element asserts itself. A great philosophical system is a work of art, a drama in which the single speeches and scenes join together, now more with logical, now more with aesthetic-ethical harmony. The systems of Aristotle, Spinoza and Schopenhauer present this character in a special degree.

This art will become harder and harder to practise. For in future it will presuppose not only that a rich material stands at one's disposal and can be shaped with constructive power, but must presuppose as well the ability to avoid dogmatizing, to maintain the significance of ideas and of constructions of thought without confusing them with absolute truths. The thing is, as LESSING said, to think gymnastically, not dogmatically. The art will be this: to unite bold development of thought with a wakeful critical consciousness. It is this side of the matter that I have particularly emphasized in my treatise *Filosofi som Kunst*. Earlier the artistic element in philosophical thinking was emphasized by SCHOPENHAUER, who however thinks mostly of the *involuntary* coming-to-be of the decisive construction of thought, whereby art's "What?" takes the place of science's "Why?" When ALBERT LANGE too called philosophical speculation an art, he was thinking rather of the *idealizing* tendency, the need to see in ideal images an expression of the highest reality.⁵⁴ — At the religious problem we shall meet a relation akin to the transition from science to art of which we have here spoken; only that there it will be a question of a completion of the view of *life*, whereas here it is a question of a completion of the view of *the world*. —

It is not, as the history of philosophy shows, at all times that the conditions are present for the exercise of the art here spoken of in the grand style. A long period of accumulation of material and of points of view, a great concentration of mind, and a thought-energy awakened by the sharpening of the problems must be present. We

⁵⁴See my *Mindre Arbejder*, p. 180–194. — On Schopenhauer and Lange see *Den nyere Filosofis Historie*. II. p. 198 f.; 501 f.

find in the history of philosophy that the systems lie gathered in certain single periods, and even bound to single places as well. The fifth and fourth centuries before Christ created the great Greek systems, and here Athens was the philosophical centre. In the seventeenth century arose the principal systems of the modern age, which came into being on the basis of the manifold material and the new points of view of the Renaissance and of the newly founded natural science; here Holland occupies a central position. At the beginning of the nineteenth century all the systems of philosophical idealism were founded in Jena, where Kant's disciples met with Goethe's disciples and with the champions of Romanticism. It is at such points of time that philosophy stands as one of the symptoms of the course of the life of the spirit; in the intervening times the more special discussion of psychological, epistemological and ethical problems goes its quieter way. —

I now go over to bringing forward the most important primal phenomena which can serve as a systematic foundation and have historically rendered such service.

3. The first fact which it lies near for philosophy to regard as the primal phenomenon in existence is this, that existence is to a great extent *intelligible* to us. We can recognize phenomena, infer from one phenomenon to another, and find continuity between them. Even though our hypotheses are really working hypotheses, and even though the old, naive concept of truth must more and more give place to the dynamic or symbolic concept of truth (see II, 2), the very fact that with our faculties and our methods we can to a certain degree orient ourselves in existence must nevertheless hang together with the essence of existence. The applicability of a method always contains a testimony about the constitution of its subject-matter. The intelligibility of existence for us points to an inner unity in its essence, which lays itself open to the day in the lawfulness that the course of phenomena presents. Our very criterion of reality consists in firm connection, and it is a natural extension of what the application of this criterion establishes when we find in existence a *unifying power* by which the single elements and events are bound together. Philosophy has a natural tendency to go back to such a principle; Plato and Giordano Bruno, Spinoza and Hegel, Fechner and Lotze worked in this direction, and even Kant, for all his great critical scruple, has borne witness to the significance of this thought. While popular thinking is inclined to seek a completion by climbing the ladder of causes so as to end in a first cause — a road that leads only to an endless series — philosophical thinking seeks its completion rather in depth than in length, bethinking itself of what the presupposition is for a rational world-picture, however hypothetical, being formed at all. *Causality* stands here as the primal phenomenon, and it is not least

for this reason that the discussion of the causal principle fills so large a place in the history of modern philosophy.

Against the monism which thus presses forward one might object that what is characteristic of the scientific understanding of phenomena is not only the assumption of an inner connection between them, but also the assumption of a manifold of elements between which the interaction takes place that the laws found by science hold for. Why not regard this manifold as the primal phenomenon, so that our metaphysics becomes pluralistic and not monistic?

The answer to this question is given in the fact that all the properties or forces with which we equip the elements of existence (whether we represent them to ourselves as material atoms or as psychical monads) are fetched from the lawlike connection in which those elements are members. The properties of things are the constant ways in which they affect one another and are affected by one another. "Force" is here the element which for us contains the ground of a change in one or more other elements. So it goes with the kindred concepts energy, capacity, individuality. All these concepts are derivative in relation to the concept of law. In the case of individuality one must here think not only of the law that holds for the individual's relation to the surrounding world, but also of the law for the relation between the individual's inner states.

In the same direction points another consideration. All knowledge begins as analysis of given intuitions (observations or recollections). If such an intuition is to be able to become an object of analysis, it must come forward as a totality embracing an aggregate of elements (parts or properties). Every act of knowledge rests on a section cut out of a greater whole: every conceptual determination (definition) is a limitation; every judgment rests on presuppositions that point out beyond the act of judging itself; every inference presupposes several premisses whose validity must often be grounded along highly different roads. We always move within a more comprehensive whole, in which the conditions must be sought for the single results we arrive at.

When thus, both in the world of reality and in the world of pure thought, the particular gets its constitution and its validity from the unitary connection in which it occurs, monism will stand as taking a more fundamental starting-point than pluralism.

Meanwhile the theory of knowledge does show that there are limits to understanding. The empiricist and the sceptic will always be able to stop the monistic metaphysician by pointing to the actual limits of knowledge. Not even the criterion of reality can we apply everywhere, and so we cannot carry through the strict difference between dream and reality. With the same right with which one infers from the possibility of rational

knowledge to a unifying power in existence, one would then, it seems, be able to infer to an irrational element in existence, to a cosmological principle which prevented the elements of existence from standing in a rationally determinable relation to one another.

From the monistic side it may again be answered that if the unifying power does not rule everywhere, that might indicate that existence itself is in becoming, in development, and that what stands for us as a lawlike connection presupposes a development in the innermost of existence which is not yet completed. It would then, from this point of view too, be *time* that conditions the irrational. And so long as thought, knowledge, is not finished but is caught up in becoming, so long *can* existence as a whole not be finished either. Thought, knowledge, is after all a part of existence! And when *it* is in becoming, there might well be more that is in becoming. It is the strange contradiction of the great rationalistic systems that even if they could explain everything else, they could not explain the striving and labouring thought whose products they are. In Plato and Spinoza, in Hegel and Boström this contradiction comes forward.

Critical monism, which holds fast to the validity of time and thereby to the incompleteness of existence as well as of knowledge, may very well build its world-view on causality and rationality as primal phenomena. It then finds in existence a struggling unifying power which through progressive development is carried out beyond what is sporadic and conflicting. Perhaps new elements are always arising which have in their turn to be worked together, and development must then always begin anew. The work of thought is hereby seen in a cosmic light: a constant connection between our methods and hypotheses and the real processes of existence is after all the goal thought sets itself (even when it exchanges the static concept of truth for the dynamic); and if thought succeeds in approaching this goal, existence will thereby have become more rational than it was before, since a new constant and harmonious relation has been realized. The thinker to whom it is given to work forward in this direction may with right say: “Ort für Ort *sind* wir im Innern” — however distant and exalted thought’s highest ideal may stand before him.

4. Can we not attempt a real, positive determination of the principle of unity which, on the hypothesis just developed, holds existence together in its innermost? — Every attempt in this direction must make use of analogy in a particularly high degree. We have here no such fundamental fact to point to as we had in setting up that hypothesis, where we could support ourselves on the connection of the problem of existence with the problem of knowledge. In a closer determination of the principle of existence, thought will be led to the most fundamental difference which the phenomena accessible to us

present: the difference between the spiritual and the material. There seems here to be a forced choice for every attempt in a cosmological direction.

How I stand, in a purely methodical and empirical way, to the problem of the relation between spirit and matter I have already developed (I, 3). But the question comes up here again from another point of view. For whether we are methodically and empirically dualists, materialists or monists, it may be asked how we think the fundamental essence of existence — what the underlying attribute is, that under which we finally — when we think our last thought — think existence.

If I lay at the base the analogy with what is extended and movable in space, then my metaphysics, and not merely my method, becomes materialistic. As a concluding hypothesis *materialism* has the advantage that it gives us the picture of a great continuity, and that we need not leave what is intuitable. Materialism is a childlike and naive conception; it is man's first philosophy. But the impression of the connection and the extent of the material world exercises a constantly overwhelming power, so that attempts in a materialistic direction come forward again and again, even if since the appearance of the critical philosophy they cannot present themselves with the dogmatic assurance of former times. They will always run aground, partly on the impossibility of tracing the spiritual back to the material, partly on the epistemological scruple that we have the material always only *as object* for consciousness, and that if materialism were right there could exist nothing *for which* the material could become object or phenomenon. HOBBS, whose thinking went strongly in a materialistic direction, nevertheless halted at precisely this consideration, that of all phenomena *the very fact that something can become a phenomenon for us* is the most important.

In contrast to materialism, *metaphysical idealism* lays at the base the analogy with spiritual phenomena. It can empirically espouse dualism (like LOTZE) or materialism (like SCHOPENHAUER) or Spinozistic monism (like LEIBNIZ, FECHNER and WUNDT). The last thought is in all forms this, that only the analogy with the spiritual states we know from ourselves gives us a key to the understanding of the innermost essence of existence. Ourselves alone do we know from within, everything else merely from without! Psychological experience already teaches us that there are many degrees and kinds of spiritual existence, and if we would use it for the interpretation of the whole of existence, of its innermost ground and its ultimate elements, then we must naturally think these series of degrees and of qualities continued without limit. This idealistic cosmology can be varied in many ways, and appears historically in many forms, conditioned especially by the other motives that determine the world-view (for example

the tendency to optimism or to pessimism, chief weight on thought or on will, and so forth). It appears already in the Indian “Upanishads”: in the doctrine that Bráhman (the world-principle) is Atman (soul).

Sometimes it is denied that it is an inference from analogy that one here relies on, and it is maintained that metaphysical idealism (or idealistic cosmology) has been reached along the strict road of logical construction, of the dialectical method. But even the proudest edifice of thought that men have believed themselves able to raise in this way, HEGEL’s system, really comes to showing that everything in existence hangs together as inwardly as the thoughts in the human mind: it is in reality the latter that is used as the basis for an analogizing, since it stands as the best example of an inward totality.⁵⁵ —

The idealistic analogizing could lead to a concluding solution only if we had means of determining in a positive way the different degrees and kinds of spiritual existence which must be found in the world if metaphysical idealism is right. But we cannot, as has already shown itself in an earlier connection (I, 3), get further here than to the negative concept of potential psychical energy. A verification of idealism is therefore impossible. And even if it could be carried through in full, there would still remain the difficulty that the material can as little be derived from the spiritual as the spiritual from the material.

But it is by no means certain that we have a forced choice between a materialistic and an idealistic solution of the problem of existence. The difference between matter and spirit is certainly a chief difference in the content of our experience; but no proof can be given that there are not in existence other fundamental forms (“attributes”) than these two. *If* the problem of existence is to be solved out of human experience, then one of those two possibilities must be chosen. But the question is whether our experience presents us with elements enough for a real solution. The richness of existence may be much greater than the possibilities of our experience. Here again it may hold that the world is great and our brain is small; here again we strike on the irrational. Could

⁵⁵See *Den nyere Filosofis Historie*. II. p. 165 f. — It is characteristic that the English thinkers who in the most recent time have worked with some of Hegel’s fundamental thoughts concede that what is in question here is analogy rather than demonstration. FRANCIS BRADLEY does not consider even the analogy defensible; the highest reality can, according to him, be called neither soul nor body, though the soul more than the body has that combination of extent and unity, that “self-consisting”, which is the mark of true reality. (*Appearance and Reality*. p. 307; 357). — In his treatise *Hegel’s treatment of the categories of the idea* (Mind 1900. p. 149 ff.) MC. TAGGART states that we have only a single example of the category “spirit”, which in Hegel acquires a theological or cosmological significance. — When one distinguishes (as A. E. TAYLOR in Mind 1900 p. 245) between two idealistic schools, one standing nearest to Leibniz and Lotze and one standing nearest to Hegel, I must maintain that it is the first of these schools that has most clearly seen the real philosophical foundation of metaphysical idealism. (That it is an analogy with the human mind that lies at the base in both cases, Taylor really concedes himself when he says that both tendencies are agreed on “the main principle that it is in mind, and nowhere else, that we are face to face with the central reality of the universe”. *ibid.*)

it be proved that the difference between spirit and matter was contradictory and not contrary, so that an absolute either-or lay before us, the problem of existence would be simpler than it is — and yet more involved than human thought, which is inclined to think itself fully equipped for religious and metaphysical speculations, has often supposed. *Critical monism*, which strives to hold fast the thought of unity without dogmatizing, must impress upon us that we here lack the conditions for a complete solution. That there is a possibility of more forms than our experience presents may indicate that the whole problem lies deeper than has been supposed. There might, for example, be a fundamental property of existence of which both spirit and matter were consequences, and the insolubility of the problem for us might come of our not knowing this fundamental property.

5. A third primal phenomenon we have when we have made our choice between subsistence and development (being and becoming). Ancient thinking was throughout inclined to hold to subsisting being; it was a philosophy of concepts, which above all sought to trace phenomena back to fixed concepts of kinds. Plato's theory of Ideas forms here the high point (compare II, 1). And it — or the same tendency of mind that has found expression in it — comes forward not seldom in more recent inquiry as well, and that not only in philosophy but everywhere in science where the battle is fought for fixed, unchangeable species or types. Even where ancient thinking assumed a development, it was particularly inclined to assume a rhythmical process which brought the same states and events with it again and again. Development as an ever forward-advancing series of changes is a modern idea, due to widened experience in the domain of history as well as of nature, but one which has also taken shape under the influence of the Persian-Christian type of religion.⁵⁶

That the concept of development comes to play so great a role for the thinking of the modern age hangs together with the causal concept's having come forward so strongly. The more the elementary concept of cause, through continued inquiry, approaches the ideal concept of cause, the more does the causal relation come to designate a continuous process in which the following members are determined by the preceding. And the concept of development comes forward as soon as it shows itself that the *direction* is essential, so that time cannot be reversed (compare II, 4). While the formal sciences, which rest on the principle of identity, can go forwards and backwards in their series of thoughts, the time-relation, and particularly the direction of time, acquires a decisive significance for the real sciences, and these thereby retain a historical character. Here

⁵⁶Compare on this type my *Religionsfilosofi*. p. 50; 121 f.; 273 f.

the very concept of event is a primal phenomenon. But in the concept of development there lies this, that not merely does something happen, but that through the series of events states are reached which have a certain character of wholeness, a manifold of elements being connected in such a way that, in spite of their difference of kind, they work in firm connection with one another and with a certain completeness over against their surroundings. Most fully, and with a great wealth of examples, HERBERT SPENCER has analysed the concept of development and maintained the combination of differentiation and concentration as its essential feature. It is by this concept as measure that we determine what — from a purely theoretical point of view — is to be called “lower” and “higher” within a changing series of states or forms.

In spite of its connection with the causal concept, the concept of development is an independent concept which cannot be derived from the general concept of cause (though it presupposes it). The causal principle would hold even if only a rhythmical oscillation took place, without progressive formation of new states of wholeness. Development stands as a fact of experience which casts light on the nature of existence. It was in and for itself quite possible that the different causal series in existence should either not meet at all, or else meet only in order to collide in a disharmonious way. Now experience shows that under certain conditions they can meet in such a way that they are combined into more composite processes and give rise to peculiar wholes. The coming-to-be of the stellar systems, the arising, the subsistence and the unfolding of life, the existence of the life of the soul and of social, historical human life are testimonies to an individualizing and totalizing tendency in existence. These phenomena constantly set inquiry the greatest problems; when we point to them here as primal phenomena, we decide nothing about whether these problems can be solved or not, but regard them merely as characteristics of existence as it once for all is. It is certainly — this must be added here — a great misunderstanding when it is often supposed that these phenomena would be more remarkable if they could *not* be explained by the help of laws that science can find. On the contrary: if a “natural” explanation of them can be found, it will be possible to infer with greater certainty than before that the individualizing and totalizing tendency lies deeply grounded in the kernel of existence.

But how deeply? Experience shows us not only, and often, a purely external and indifferent relation between different causal series, but also, and often, collisions between them, or at any rate disharmonious relations which hamper or hinder the arising or the subsistence of individualities and totalities. And after development follows dissolution, and the question becomes whether through the rhythmical alternation of these processes

a progressive course of development can be traced, or whether we must halt at the ancient thought of rhythm as the last thought.

Here the irrational comes forward again, and it comes forward here in a sharper form than at the earlier points where we met it; it designates here what hampers and dissolves, the tendency to remain standing at, or to return to, the most elementary forms of existence. Existence gives us here, the more inquiry penetrates into it, the picture of a struggle — the great struggle which all forms of existence that bear the stamp of individuality and of totality must wage in order to subsist. The struggle itself is again two-sided: it can be a means to development, but it can also be a road to destruction; and which of these possibilities is the preponderant one?

At the same time there shows itself here still more clearly than at the earlier points the impossibility of forming an absolutely completed concept of existence as a whole: if the conflict between elements and totalities is an *essential* feature of existence, then it is a feature that can always be found only in limited sections of existence, not in existence regarded as a whole, since an absolute whole can meet no outward resistance, can be engaged in no struggle. We must content ourselves with saying that wherever we come upon existence, we find that within it such a conflict between elements and totalities goes on. The great question is whether through this conflict it is the elements or the totalities (the solar systems, the organisms, the souls, the societies) that carry off the victory. Empirically we stand here in the midst of the great world-process, and can get no further than to maintain that it is no purely subjective measure we apply when we designate one form of existence as higher or lower than another, since development is a phenomenon characteristic of all the existence we know — a primal phenomenon. —

By the result at which we thus end in the matter of the problem of existence, a natural transition to the last problem, the problem of valuation, becomes possible.

If existence were finished, harmonious and unchangeable, no ethics would be possible. All ethics demands a work performed; but there would be no room for any work if everything consisted in eternal and actual completedness. Precisely this, that all individualities and totalities must struggle in order to subsist — that there are constantly disharmonies to overcome, sporadic tendencies to connect — precisely this makes an ethics possible. For ethics investigates the principles of a carrying-further of the individualizing and totalizing tendency which experience exhibits in existence. Ethics rests on a self-contradiction if the course of the world does not partly proceed, or cannot partly proceed, through human will, just as it partly proceeds or can proceed through human thought-work (III, 5). Ethics then takes up the problem of continuity where the first

three problems let it go.

Still more clearly does the religious problem find a point of attachment in the foregoing consideration, since it — as I shall attempt to show — concerns the continuity of values under the struggle of existence.

IV. The Problem of Valuation

1. That the problem of valuation is set up as a separate problem presupposes a distinction between understanding and valuation which has not always been acknowledged, and which has not yet made its way through. There has been an inclination in philosophy to let the two run into one: thus Plato's "Ideas" and Spinoza's "substance" are at once an expression of the two thinkers' understanding of existence and of their admiration for it. In any case men have been inclined to let valuation be a mere consequence of understanding, while conversely mystical and theological tendencies have been inclined to let understanding be dependent on valuation. Wholly independent of one another they certainly are not: understanding has its value, and would not be possible if that were not so, since men would not otherwise strive after it — for some, indeed, the joy of knowledge is the highest of all; and valuation can become an object of psychological and historical understanding, since all value rests on the relation of experiences and states to life at its various levels, to life's subsistence and its development. The problem of valuation will therefore also show itself to present an analogy to the three first, theoretical problems. It is in the domain of valuation, as in the domain of personality, of knowledge and of existence, the principle of continuity that leads to the decisive problems.

Value belongs to everything that brings a satisfaction with it or relieves a need. Sometimes it is only through a satisfaction's arising that we become aware that there has been a gap in our existence; sometimes this gap is felt beforehand and causes a want or calls forth impulse and desire. When what is valuable does not immediately fall to our lot, we make it an *end* and seek means of attaining it. Whatever stands as a means of attaining something immediately valuable acquires a mediate value for us. In our values and our ends the innermost essence of our feeling and our will makes itself known. Just as the concept of end depends on the concept of value, so the concept of *norm* depends in turn on the concept of end: the norm is the rule for the activity necessary to the attainment of the end.⁵⁷ — It had a fateful influence on the treatment

⁵⁷In my *Etik* (2nd ed., p. 21) I did not enter more closely into the relation between the concepts value, end and norm. See instead *Det psykologiske Grundlag for logiske Domme*. p. 47 (compare p. 65) and my

of the problem of valuation that IMMANUEL KANT wanted to reverse the relation and to derive the concepts of end and of value from the concept of norm ("law"). That is a psychological impossibility.

Now experience shows that for different individuals, and for the same individual at different times, different values hold. I have already mentioned the value of knowledge; besides it there are to be mentioned especially the values attached to the impulse of self-preservation and to the subsistence and development of life in larger or smaller circles — to movement and activity — to images (those of reality or of fantasy), and so forth. If a comparison is to be instituted between different values — and every conscious valuation consists in such a comparison — then a *fundamental value* must be presupposed, from which the order of rank of the different values can be fixed. A determinate value must be laid at the base and furnish the measure for all other values, if consistent thinking and discussion is to become possible in the domain of values. From a given fundamental value there can then — on the presupposition of sufficient experience — be constructed a system of valuation in which every single value gets its place according to its relation to the fundamental value. Thinking, "practical reason", is needed in order to determine this relation, and also in order to find means and ways of producing or finding out the several values under certain given conditions. But on bare ground no valuation (or "revaluation") is possible. The concept of fundamental value is, within the problem of valuation (that is, in ethics and the philosophy of religion), what the concept of primal phenomenon is within the problem of existence (that is, in metaphysics). —

The problem of valuation divides into two problems, the ethical and the religious. Ethical valuation concerns human actions, qualities and orderings of life (institutions); religious valuation reaches further and values existence according to the fate of values in the world of reality. Within both problems the relation between continuity and discontinuity will — as at the earlier problems — show itself to be of decisive significance.

A. The Ethical Problem

a. Ethical Work

2. There is, as the discussion of the problem of existence has shown, room for a work by which existence is developed further. Such a work is performed in all human culture, but quite particularly in ethical striving. This may be designated as a striving to bring forth greater continuity, partly within the single personality, partly between the different

personalities among themselves. The measure of ethical striving, and so the principle of ethical valuation, will show itself to be determined by the principle of continuity.

Personality presupposes continuity, and this is in turn conditioned by there being one fundamental value which determines the value of the several moments, periods of life, capacities and impulses. Development into a true personality presupposes a striving out beyond what is momentary and what is sporadic — an overcoming of the tendency of the single moment or the single need to isolation and sole dominion. The thing is to work the several moments and elements harmoniously into the life of personality as a whole. Here is a work to be done, a struggle to be come through, which in different individuals demands different degrees of energy.

There are ethical standpoints which assert the right of the several moments and impulses, of the successive and simultaneous differences, over against the whole of life. But this does not conflict with the measure's being fetched from the principle of continuity. Such standpoints may be warranted in so far as they demand that the subordination of moments and special impulses to the whole of life be grounded. Continuity does not mean absence of difference, but the ordering of the differences into a graded series. Life as a whole can always be called to account by the single element of life. It will always stand as an imperfection when a moment, a period of life, a capacity, an impulse is made a *mere* means without acquiring independent value. The art will be to let them have immediate and mediate value at once. — And in so far as the ethics of the moment or of the speciality appears in absolute form and proclaims the sovereignty of the single moment or element, one will often find at the base the presupposition that life as a whole is represented by, or is concentrated in, the single moment or the single need, so that all other considerations pale. So it is in the great moments of self-sacrifice, where a man — as ARISTOTLE says — “would rather perform one great and beautiful action than many small ones”; and so it is where a man sees the calling of his whole life in the special development of a single capacity. Or one believes in a harmony between all the different moments of life, so that a complete absorption in each of them becomes possible without the satisfaction of the single moment depriving the other moments of anything. So ARISTIPPUS seems to have thought of it. Or the highest is placed in a hovering above the several moments and occupations, with the possibility of alighting at will, and without being bound, on the single point for a while. So in the “aesthetic” view of life depicted by S. KIERKEGAARD in Part I of *Enten-Eller*. Under all these forms there plainly lies a regard for the whole of life in the background.

If a contradiction is experienced between what is momentary or sporadic and the

need for wholeness, there will arise a more or less conscious and a more or less energetic striving to develop the personality into a work of art, in which every single moment and every single capacity gets its place and comes into its own. No element in the personal life is then regarded as a mere means, but always at the same time as an end. What is a means or a point of transition for the development of the personality shall so far as possible have immediate value itself. It is *the unforgettable fundamental thought of Greek (Platonic-Aristotelian) ethics* concerning the harmonious unfolding of the life of the soul as the great task — a fundamental thought which has its significance quite particularly under modern culture's tendency to isolate or to mechanize the several elements of life. ROUSSEAU and SCHILLER have renewed this ancient fundamental thought — against modern labour-helotism, and against dalliance and decadence.

But the problem of continuity comes up again, and in a very sharp form, at the question of the possibility that the single personality can form a completed world for itself, and whether that would be the most valuable thing. Not only must the individual always enter into relations of interaction with other personalities in order to obtain means for his own development. There is also a need for self-surrender, which can appear in different forms and which can lead to ascribing immediate value to other personalities. The individual is thereby drawn into the great realm of personalities; and just as the question before was how far the several elements of life could be harmoniously ordered into the whole of life of the single personality, so it now becomes a question how far the several personalities can develop independently and yet in harmony with one another, so that there can subsist *a social whole of life* in analogy with the individual whole of life. It is continuity within continuity that is here striven towards. The measure of a society's perfection will — in virtue of the principle of continuity, which here shows itself plainly in its connection with the principle of welfare — be this: in what degree every personal being is placed and treated in such a way that it does not stand as a mere means, but always at the same time as an end. It is KANT's famous proposition, with a different grounding from the one he gave.⁵⁸ Stoic and Christian ethics and the modern social and political development lead in the same direction. The declaration of the "rights of man" (whether we regard it as a symptom or as a programme) rests on this presupposition, and the sting in the social question has come about through this presupposition not having been satisfied. In the discussion of more special ethical and legal-philosophical questions too — for example on the grounding of monogamy and on the development of the penal system — this principle furnishes the measure. —

⁵⁸Compare my *Etik.* 2, p. 144–151.

Ethical work thus shows itself as a peculiar continuation of the great process of existence. But the whole train of thought by which I have sought to ground this point of view runs into difficulties which sharpen the ethical problem. We meet here again the irrational.

b. The Rationality of Ethical Valuation

3. Ethics would be a more perfect science than it is if the carrying through of the principle of continuity did not run into great difficulties.

Every ethical argument holds only on the acknowledgement of a determinate fundamental value which determines all special valuations. One may take one's stand on the standpoint of the single moment or the single impulse, or on that of the isolated personality, or on that of the family, or the class, or the state, or humanity. The question is whether all such standpoints can really be brought into harmony with one another, as has been assumed above; and the question is especially whether it is possible, by way of argument, to bring a man who consistently and with the requisite ruthlessness holds fast to one of these standpoints over to one of the others. Between the fundamental value and the special values a rational relation can perhaps be exhibited, so that one who avows a fundamental value, and is sufficiently acquainted with the actual conditions under which it is to be held fast, can be logically compelled to concede the inferences one draws from the fundamental value. Thus self-assertion and self-surrender have each their logic, and so too have the feeling for family and the feeling for nation. But what when it is a question of *transition from one fundamental value to another*? Here inner consistency is not enough. Consistency can only unfold and bring to consciousness what — under certain given, actual conditions — follows from the fundamental value. SOCRATES became the founder of ethics by his demand for self-knowledge, which was in reality a demand to become clear about one's fundamental value and about what follows consistently from it. But how the fundamental value is procured, and whether there are not in fact different values which claim to be fundamental values, he did not discuss more closely.

A fundamental value forms itself by a psychological and historical process which presupposes more operative elements than logical consistency and knowledge of facts. Through displacement of motive and of value⁵⁹ one fundamental value can pass over into another; but *while* such a fundamental change is going on in the domain of the life of feeling and will, it will be of no use to argue from the presuppositions which only come

⁵⁹Compare my *Psykologi*. VI. B., 1–5. — *Etik* 2 XIII., 4.

to be present when the process is finished. In education one cannot without more ado argue with the child from the presuppositions of adults; the child has precisely first to acquire those presuppositions. And so it goes too with the great process of education that goes on in history. *During* the education and the development there are of course other fundamental motives at work than those which only become the result of the whole process. He who is being educated will therefore not be able rationally to understand the system by which he is being educated. “Wer kann der Raupe, die im Staube kriecht, von ihrem künftigen Futter sprechen?”

There does not, now, take place an evenly advancing education in history from primitive fundamental values to higher ones (that is to say: to such as those who have acquired them call higher in comparison with the primitive). History is the great battlefield of fundamental values. Here individual stands against individual, individual and society against each other, and society stands against society, and it is often only after violent conflicts that a new fundamental value can be fixed in the minds of men. An example of this is the way in which the conquests of Alexander and of the Romans prepared the way for the development of the general feeling of humanity. During this great struggle ethical thinking can play a part only indirectly: by drawing the determinate consequences of each single position, and by illuminating the significance of those consequences by as many experiences as possible — that is, by furthering self-knowledge in the spirit of Socrates. In the struggle itself, or in the education, what is called for is an art; science does not suffice, however significant a contribution it may often make.

But we do not abandon the principle of continuity because we stand at the boundary of scientific ethics. There, where it is not possible to pursue continuity further in purely ethical form, we seek to trace it out in psychological and historical form. Ethics passes at this point over into psychology and into sociology.

While I have sought in my *Etik* to show that justice, conceived as an inner harmonious relation between self-assertion and self-surrender, and also between feeling and thinking, is the highest quality of character, this has been disputed by philosophical thinkers who maintain that in self-assertion and in self-surrender such different tendencies (or, as I should say, fundamental values) make themselves known that it is not possible to unite them in one embracing concept.⁶⁰ And in popular literature self-surrender (altruism) has been asserted just as ruthlessly by LEO TOLSTOY as self-assertion has by FRIEDRICH NIETZSCHE. Here we have an example of a strife between fundamental values that press

⁶⁰FRANCIS BRADLEY: *Appearance and Reality*. p. 414–418. — A. E. TAYLOR: *The Problem of Conduct*. London 1901. p. 201.

forward at the same time. There will probably always, in actual human life, be oscillations between these two opposite poles. In the race as a whole both self-assertion and self-surrender have their function to exercise. But the valuation of the single oscillation in the one or the other direction will always depend on whether it leads forward in the direction of an ordering of life within which every single personality can develop as peculiarly and independently as possible, in such a way that precisely thereby it renders the best help to the equally peculiar and independent development of others. In individuals self-assertion and self-surrender will stand in highly different relations to one another, and the harmony between the two, in which on my view justice consists, will therefore appear with a highly different timbre in different persons and at different times. But that too belongs to the richness of life — and to its continuity. The manifoldness of the nuances conditions the connection and makes it different from a dead uniformity, which has no value either from an ethical point of view or from any other.

4. Another difficulty, one connected with discontinuity, arises from the fact that the inferences drawn under certain conditions from a fundamental value have to be applied to *individuals who come with highly different inner and outer conditions* for satisfying the demand which such an inference grounds. Aptitudes and impulses are not the same in all individuals, either in kind or in degree. One and the same demand addressed to different individuals therefore lays on them a highly different ethical work. The starting-point and the initial speed are different. Some individuals are perhaps already on the way to an involuntary performance of the content of the demand before the demand becomes conscious to them; others must struggle laboriously to lay the first foundation. The law, the demand, must therefore be set differently for different persons if it is really to become the same for all. Each must be taxed according to ability. A thoroughgoing individualization of the ethical demand must be undertaken, if ethics itself is not to sin against the proposition that personality is always an end and never merely a means. The ethical demand shall not be an abstract or external command, but shall answer to the ethical possibilities in the single personality and be able to develop them further. Legislation and pedagogy cannot here be kept absolutely apart from one another. But this makes ethical decisions difficult in the several cases.⁶¹ Here again the world — the world of personalities — is great, and our brain is small. Ethical thinking cannot formulate a law that can be applied without more ado in the manifoldness of life. And yet we must assume that in each single case only one single decision can be the right one (Eng ist die Welt, und das Gehirn ist weit). Our ethical thoughts circle about the narrow way

⁶¹Compare my *Etik.* 2, p. 70–73.

that leads to ethical truth, and work their way forward to it through a constant struggle with the irrational, which they seek to overcome by finding as great an approximation as possible. Here, as within the problem of knowledge, we do not end with an absolute conclusion, but hear an *Excelsior!* — even where our thought has strained itself to the utmost.

B. The Religious Problem

5. The very fact that a religious problem exists already shows the significance of continuity. For that religion has become problematic hangs together with a division of labour, a differentiation, having set in within the domain of the spiritual life. In religion's classical ages it stands as the one concentrated form in which all the requirements of the life of the spirit are satisfied; religion is as such not merely help and comfort, but is poetry, morality and science, or it is at any rate able to take these domains up into itself in an organic way. After these various domains have emancipated themselves, and each develops according to its own laws, the question arises how far the life of the spirit — as regards value — has preserved its continuity through that transition from concentration to differentiation. That a psychological and a historical continuity is present there is no reason to doubt; but has not value been lost for human life by that transition?

The answers to this question sound highly different. Some are content that the same dogmas are still taught as in the old days. From another side it is maintained, on the contrary, that this dogmatic continuity means nothing: the essential thing is whether life is led in the same way — whether there is ethical continuity — whether “the Christianity of the New Testament” still exists. And finally there are those who hold that with the division of labour in cultural history the age of religion is past, and that this means no loss of value but on the contrary a gain for the life of the spirit. Besides these principal standpoints, which can themselves appear in different nuances, there is a series of different and less characteristic standpoints.

The answering of the question depends of course on what one takes to be the essential thing in religion. Continuity cannot appear in such features as separate the different religions and religious standpoints from one another; one must seek out the most deeply moving tendencies, which can reveal themselves under highly different forms, and which may perhaps also continue to work after the division of labour in cultural history has come into force. This is the task of the philosophy of religion, and it seeks to solve it especially along two roads. First, it institutes a comparison between religion and other

sides of the life of the spirit in order to find in which region of the life of the spirit religion belongs; it seeks to determine religion's psychological place. And secondly, it institutes a comparison between the most important historical forms of religion in order to see whether the psychological determination found wins confirmation in experience.⁶²

In both investigations we shall have occasion to see that it is not only the principle of continuity that leads to the posing of the religious problem, but that the very concept of religion hangs very closely together with this principle.

6. The religious feeling can be placed in a clear and simple relation to the other feelings when it is conceived as determined by the experiences man makes concerning the fate of values in the world of reality. It is thereby seen at once in its difference from and in its connection with the other feelings. To every feeling there corresponds a value. The feeling of life, the intellectual, the aesthetic and the ethical feeling thus make known different kinds of value. The maintenance and development of life, truth, beauty and goodness are values man can have a part in without religious feeling. But when it shows itself that life, truth, beauty and goodness must struggle in order to subsist in the world, there arises a peculiar feeling which is no longer determined by those values in and for themselves, but by their subsistence, their advance or their decline in existence, so far as man is able to survey it. The experiences man makes about this will be able to give rise to a need to believe that the values subsist, even where they no longer make themselves known in the reality man can survey, or even if they no longer appear in the same forms as hitherto. Man will be most inclined to assume that the value which subsists is the same one he has hitherto had a part in; but he may develop to the point of holding fast that the empirical values must perish in order that a higher and more comprehensive connection of values may be satisfied. From men's representations of a world of gods and of a life beyond, one comes to know which values have lain nearest their hearts, and in what degree they have reconciled themselves to the thought that the empirical values must undergo a more or less essential metamorphosis if a subsistence of value is to be possible. In the end what subsists is something that no eye has seen and no ear has heard. Religious faith asserts a continuity in the domain of values beyond every possible experience. This continuity of value may be of different content and extent. But there will be a tendency to extend it beyond the human world and to conceive existence as a home for the development and the subsistence of values. This religious

⁶²In my *Religionsphilosophi* I have — for a more comprehensive illumination of the problem — instituted, besides the psychological-historical investigation, an epistemological and an ethical one as well. Here, in this shorter presentation, I keep especially to the psychological-historical investigation, and only indicate quite briefly, here and there, the epistemological and the ethical points of view.

faith in continuity forms an analogy to the intellectual presupposition of the rationality of existence (III), and the difficulties it has to struggle with are analogous too.

Religious faith will have in particular a certain likeness to (and a certain sympathy with) metaphysical idealism (III, 4). But it does not stand or fall with it. The essential thing in religious faith is not the intellectual satisfaction it can afford. The circle of representations in which religion is expressed does not belong to its innermost psychological essence. What is central in religion is — if the psychological determination given is correct — an interest of feeling and of will. Intellectually we ask only about recognition (identity), consequence (rationality) and causal connection (causality) (II, 1). Faith is always only an *object* for science, never itself science. It arises through the harmonious or disharmonious relation between the picture of reality which knowledge gives man, and the values which stand for man as the highest.

Possibly the difference we are compelled to make between value and reality holds only from a human standpoint. But — another standpoint we cannot, once for all, take up.

The faith in the subsistence of value can itself acquire value by keeping courage up in the struggle for life, and by spurring us to find new values as equivalents for the values that disappear. Even one who assumes that the age of religion is past — an assumption which, if it is to be more than an assertion or a wish, must be grounded epistemologically, psychologically and ethically — even he will surely feel a need to find equivalents for the loss of value suffered through religion's disappearance. In this sense a religious problem exists also for one who holds that religion's kernel disappears along with its shell. For religion, by asserting the subsistence of value, acquires value itself, value in the second power. And it designates in particular one of the most concentrated forms of spiritual life that experience offers: it stands, where it is genuine and original, as the collected result of man's life of feeling and will, and it gets itself expression in the mightiest and noblest images that man's life of representation commands. Probably this concentrated life-process will always express itself under new forms — whether or not one goes on using the word religion.

7. A confirmation of the characterization given of what is essential in religion may be sought in the fact that the differences between the religions, or between the religious standpoints among themselves, can simply and easily be explained from it. These differences rest partly on differences in the values in whose subsistence there is belief, partly on differences in the conception of reality that lies at the base, partly on differences in experience concerning the relation between the values and reality.

The differences with respect to *the values* appear in the historical religions in what is regarded as the task of the gods — what one thinks of the gods as struggling for — or, to use an expression of PLATO's, in *that which makes God divine*.⁶³ It may be the purely physical maintenance of life and the enjoyments attached to it, or it may be the good, the beautiful and the true. Historians of religion emphasize the transition from nature-religion to ethical religion as the most significant within the history of religion; but this transition rests precisely on a transition from the more elementary values to the ideal values as that which lies at the base of the relation to the gods — what the gods guard, and to which they themselves owe their divinity. At the same time the more external relation between the powers man believes himself to stand in relation to, and that which those powers guard, falls away, and the gods themselves become more immediate representatives of the valuable which they uphold — indeed become one with it. An ever more inward connection between the religious and the ethical can be traced in religion's development, and religion tends more and more to become a projection of the ethical.

When *the picture of reality* is altered by advancing knowledge, there occurs a no less decisive alteration in religion's character than when the values lying at the base are altered. The most conspicuous religious crises are due to a new conception of reality. Religion has here on the whole behaved receptively towards the knowledge developed along another road, but has afterwards sought to take it up and to use it as a foundation for its images. Thus animism, the knowledge of the stars, the doctrine of the transmigration of souls, Indian and Greek philosophy, Copernicanism and more recent natural science, speculative and critical philosophy have successively intervened — now as welcome forms for the religious life of representation, now as widening impulses, now as hostile tendencies.

But the decisive thing will always be *the relation between the values and reality*. According as this relation appears in man's experience of life as harmonious or as disharmonious, religion will acquire a different character; and in the highest religions both life's disharmonies and its harmonies will find their place — there the religious mood will first reach height and firmness when it has worked its way through conflict and suffering to blessedness. The strength of the disharmonies then becomes only a testimony to the strength of the victorious harmony. But to this comes another relation. Religion will acquire a different character according as what is highest in value is thought as subsisting eternally, raised above all becoming and change, so that life in time is in

⁶³PLATO: *Phaedrus*. p. 249 C (πρὸς οἷσπερ θεὸς ὧν θεϊὸς ἐστὶ). (More exactly translated: "that in which it is God's divinity to live".)

the end an illusion, or as the valuable itself unfolding in time and struggling for its subsistence in the course of the changes. On this difference rests the opposition between the Indian-Greek type of religion and the Persian-Christian type.

It seems then that it is no alien measure we apply to the religions when we characterize and value them according to the way in which, and the degree to which, the principle of the subsistence of value appears in them. It is a measure that naturally presents itself for a comparative consideration of the history of the religions and of religious standpoints.

The second of the types of religion named stands in history as the victorious one. It has certainly not reached its definitive form. If it comes to take on new forms, that will be due, as in the past, to prophetic personalities. They alone are able to weave the Godhead a new garment.

For the philosopher the essential interest is to exhibit that so long as a religious problem subsists, it hangs closely together partly with the need for continuity, partly with the time-relation, partly with the ethical values. The kinship and the inner connection between the great problems of the human spirit, the practical as well as the theoretical, here come plainly forward.

And at all the problems we end — when we see the opposition in its sharpest form — in the eternal strife, a strife in which the spiritual powers of men shall ever anew be tried. But because we cannot solve the great problems, we can nevertheless see the way that leads forward in such a manner that the right both of thought and of life is maintained. The insolubility of the problems means really only this: that however far we get in our inquiry and our thinking, there form themselves for us ever new horizons, new goals and new tasks.